

CareerConnectBD: A Web-Based Career and Recruitment Management Platform

Course Code: CSE-400

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Student's Declaration

We hereby declare that the project entitled “**CareerConnectBD: A Web-Based Career and Recruitment Management Platform**” This is an original work submitted to the Department of Computer Science and Engineering, Sylhet International University (SIU) under the supervision of **Md. Umor Faruk Jahangir**, Assistant Professor and Head, Department of Computer Science and Engineering, Sylhet International University (SIU) in partial fulfillment of the requirement for the Degree of Bachelor of Science in Computer Science and Engineering.

We also state that this work has not been submitted earlier for the degree, diploma or certificate of any University or other body and that all information sources used in this research work have been duly referenced.

We declare that this work is in line with the academic integrity policy of Sylhet International University. Does not contain plagiarism or fabrication, falsification, or other academic dishonesty. Only tools based on artificial intelligence were found to be limited to language refinement, formatting, coding assistance and organisation of literature. All research design, execution, analysis, interpretation and conclusions are the sole responsibility of us and we are responsible for the originality, accuracy and integrity of this project.

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Bonafide Certificate

This is to certify that the project entitled “CareerConnectBD: A Web-Based Career and Recruitment Management Platform”, submitted by **Md. Tareq Miah(Student ID: 223101017)**, **Asif Mahmud Khan(Student ID: 223101005)**, and **Sunia Akther(Student ID: 223101015)** to the Department of Computer Science and Engineering, Sylhet International University (SIU), in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering, is a record of original research work carried out by them under my supervision and guidance.

To the best of my knowledge, this project has not been submitted elsewhere for the award of any degree, diploma, or certificate. I hereby recommend this project for examination and evaluation.

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Abstract

CareerConnectBD is an online career and recruitment management system designed to offer an integrated career management system for three user groups: Job Seeker, Employer, and Administrator. The system carries major challenges of broken workflows in the fragmented job search, candidate management, resume preparation and recruitment tracking processes to a single role-based application.

The platform was developed with the help of the following technologies: Laravel 13, PHP 8.4, MySQL, Laravel Breeze (Authentication), Laravel Eloquent ORM (Data Access). The frontend uses Vue.js for the interactive part and components, Bootstrap 5 for responsive design and, when needed, Laravel Blade for server-rendered views and layout. For Frontend Asset Development and Production Builds, Vite is used. The overall system is based on MVC design, uses middleware, validates and implements role-based access control to secure role-specific resources.

Job Seekers may create profiles (including education, experience and skills), keep resumes (which includes upload resume and resume builder), search and filter for jobs, save jobs, apply for jobs and monitor the status of their applications. The implementation allows employers to set information, publish and keep up to date job postings, see the applicants, update the status of the applications, and manage the recruitment steps for interviews. Admins can track and see important platform data, manage users and operational content, and oversee support and feedback modules (if applicable).

CareerConnectBD provides a working notification workflow based on persistent notifications from the database for recruitment and platform related events like job applications, job status changes, support related changes etc. The system is implemented and tested in a local development environment and its design allows it to be deployed in a compatible production hosting environment.

Keywords: CareerConnectBD, Job Portal, Recruitment Management, Laravel, Vue.js, Applicant Tracking, Role Based Access Control, Notifications, Resume Management.

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Introduction

1.1 Background of the Project

Digital technologies have drastically changed the recruitment process for organisations and also the way people look for a job. Digital technologies have revolutionized the way organisations recruit employees, and the way people seek employment. Online career sites have opened possibilities for people to find jobs and for employers to market themselves to people via the internet. But, the hiring procedure is not just about putting up the job notices and receiving applications. Keeping professional profiles, preparing resumes, researching job opportunities, checking follow-up and monitoring, and communicating with employers are some of the job search requirements that job seekers need to fulfill. Likewise, employers need effective tools to help them keep their company information under control, post job ads, take a look at candidates, assess job seekers, and coordinate recruitment efforts.

CareerConnectBD is designed as a web-based career and recruitment management platform that combines these activities into one role based management system. The platform is created based on three main user types: Job Seeker, Employer, and Administrator. Dedicated functions and access to system resources are assigned to each role, based on its responsibilities.

The system offers facilities to the Job Seekers to maintain their career profile which consists of their education, past experience, and skills. Users can upload, edit and manage their resumes, can build resumes with a built-in resume builder, search and filter available jobs, save their favorite jobs, submit applications and track their activity with the application.

Through implementation, employers are able to build and manage company profiles, post and manage job openings, review applications, access information about candidates and resumes, update application statuses, and manage the activities involved in an interview.

The system also offers an administrative environment that allows for centralized management of users and significant activities of the platform. Administrative features aid in managing the platform, platform monitoring, supporting activities and feedback management (if available). Additionally, CareerConnectBD has a notification workflow that facilitates communication between Job Seekers, Employers, and Administrators.

CareerConnectBD is developed with the Laravel PHP 8.4 framework which includes PHP 8.4, MySQL, Laravel Breeze (authentication), and Eloquent ORM (database interactions).

The frontend features Vue.js for interactive features, Bootstrap 5 for responsive design, and Laravel Blade for server-rendered views and layouts (if applicable). The application uses middleware, validation and role-based authorization to make it more maintainable and secure, and it follows the MVC pattern.

1.2 Problem Statement

There are a number of hurdles that can be encountered along the way of any job search or a job offer. Job seekers must plan their professional data, keep resumes, look for suitable job opportunities, apply for jobs, and track the process of recruitment. If they're not coordinated via a unified system, the job search process can become disjointed and hard to follow.

Another obstacle is finding appropriate employment opportunities. It is hard for a job seeker to gauge the extent to which a job aligns with his or her skills, education, experience, preferences, and resume information. In the same way, an employer can get more than one application for a job and need to have an organized way of evaluating candidates and deciding if they are qualified for the job by the relevant requirements.

Resume preparation is another important aspect of career development. Candidates who are looking for a job might have to keep multiple resumes uploaded or develop a professional resume on their own without depending on any external sources. Besides, if you have incomplete profiles and a poorly organized resume information, it can have a negative impact on your applications for jobs.

Employers also need an arranged recruitment procedure. They should have to keep track of company information, set up and update job posts, screen applicants, be able to view resumes, update application status, and coordinate interview activities. If they are not managed centrally, these processes can prove to be cumbersome.

As far as the administration is concerned, there have to be proper mechanisms to manage users and keep a track of key activities on the system. There is also a need for the communication between various participants. Timely notifications to appropriate users may be required for job applications, application status changes, interview activities, support requests and administrative responses.

Thus, there is a need for a career and recruitment management platform that can be integrated and have management of job seeker profile, resume services, job discovery, application tracking, management of employer recruitment, rule-based decision support, administration, and communication all in one system. CareerConnectBD is designed with a modular, secure and role-based web application to meet these requirements.

1.3 Goals and Objectives of the Project

The primary goal of CareerConnectBD is to design and develop an integrated web-based platform that connects job seekers and employers while providing administrative control over the overall system. The platform aims to simplify career management and recruitment activities through structured workflows, role-based access, and rule-based decision-support features.

The major objectives of the project are as follows:

- To develop a secure role-based authentication and authorization system for Job Seekers, Employers, and Administrators.
- To enable job seekers to create and maintain structured career profiles containing personal information, education, professional experience, and skills.
- To provide resume management facilities that allow job seekers to upload, maintain, preview, and download resume files.
- To provide a built-in resume builder that allows job seekers to create, edit, preview, print, and generate professional resumes through the platform.
- To enable job seekers to search, filter, view, and save available job opportunities.
- To allow job seekers to apply for suitable jobs using their resume information and track submitted applications.
- To enable employers to create and maintain company profiles and manage organizational information.
- To provide employers with facilities for creating, editing, duplicating, deleting, restoring, and managing job postings.
- To provide an organized applicant management process through which employers can review candidates, access candidate profiles and resumes, and update application statuses.
- To support interview-related recruitment activities and communication between employers and job seekers.
- To implement rule-based candidate-job matching and candidate ranking mechanisms based on relevant profile and job information.
- To provide rule-based resume readiness and content analysis to identify resume quality factors, relevant sections, and improvement opportunities.
- To provide personalized rule-based job recommendations using available candidate profile and job-related information.
- To provide an administrative panel for centralized management and oversight of major platform resources and activities.
- To provide notification and communication mechanisms for important events such as job applications, application status changes, interview activities, support interactions, and administrative communications.
- To provide support ticket and feedback mechanisms for communication between platform users and administrators.
- To apply structured software engineering practices, including MVC architecture, service-layer separation, validation, middleware, policies, testing, and version control throughout the development process.

1.4 Scope of the Work

CareerConnectBD aims to create a web-based job seeking and recruitment management website for three main users namely Job Seeker, Employer and Administrator. It offers different interfaces and functions for various roles within the system, while ensuring that the system is centrally managed with regard to authentication, authorization and data processing.

The scope includes profile management, education management, experience management, skill management, uploading / managing resumes, support for multiple resumes, default resume selection, resume preview / download, and profile completion monitoring for the Job Seekers. The system also features a resume builder, which allows users to build and keep their resumes organized, and offers preview, print and download capabilities.

Public job listings, job details, search and filters, and saved job functionality are amongst the job discovery functions. Job seekers can submit job applications, choose relevant resumes (if appropriate), enter information relevant to the job application, avoid duplicate submission of an application, and track their application activity once their applications have been authenticated.

The scope for Employers' includes creating Company profile and Company logo management. Employers can create a job posting and manage it, including editing, deleting, soft deleting, restoring, and duplicating jobs. Employers may view applications submitted, view candidate details, preview or download resumes, assess candidate suitability and update application statuses. The recruitment workflow can also be used for activities related to interviews.

The administrative scope encompasses a unified admin dashboard and administration of essential resources and activities on the platform. It also supports the handling of support tickets and the management of feedback, based on the modules that have been implemented in the administrative part of the system.

Database-driven notifications of relevant business events are included within the communication, such as submission of job applications, updates on the status of job applications, activities related to interviews, and support/feedback notifications (if applicable). The basic notification flow is still database-driven, but if applicable, can be enhanced with optional realtime broadcasting.

The project is centered on the design, implementation, integration and testing of the web application. Large-scale production infrastructure, native mobile applications and advanced artificial-intelligence integration (such as trained machine-learning-based matching and recommendation) will be considered as future enhancement areas, and will not be included in the current implementation.

1.5 System Overview

CareerConnectBD is a role-based system architecture, where each user is granted different capabilities based on their particular role. The three main functions of the platform are: Job Seeker, Employer, and Administrator.

The Job Seeker module is about creating a career profile and job-seeking. A Job Seeker

may have access to and maintain information regarding their education, work background, skills and resume. The user can view available jobs, job details, save jobs of interest, apply for jobs, and check the status of their applications. A resume builder and resume management offer extra resources for career documents.

The Employer module is dedicated to the management of recruitment and candidate. An Employer can be linked to information regarding the company and can use the platform to manage its job postings. The Job Seekers can add their applications to the recruitment workflow for review. Employers have access to relevant candidate information and resumes, the ability to update the status of the applications and can do interview related actions supported by the system.

Centralized management and oversight is available via the Administrator module. Administrators have access to an administrative dashboard from which they can manage or monitor key platform resources based on the administration features implemented, including support and feedback modules, if implemented.

The system also features a communication and notification component that enables the mapping of critical business activities with the relevant users. For instance, Employers are notified about applications being made, Job Seekers are notified when the status of their applications change or when they receive actions related to the application or interview process, and Administrators receive relevant support and/or feedback information. The key notification method is persistent database notifications, and there is also the optional real-time broadcasting, which can be added for enhanced delivery if available.

Technically CareerConnectBD is a web application that is built using Laravel. Requests are handled on well-defined paths and are secured with authentication, middleware and authorization. Services are used as needed to wrap business logic; controllers are responsible for coordinating handling of requests. Eloquent Models deal with interactions with the MySQL database. The frontend uses Blade to generate templates, Bootstrap 5 for styling and Vue.js for interactive user interfaces (where applicable).

All modules are combined to create the entire system as a career ecosystem, where job seekers can build their professional profiles and find career opportunities, employers can process the recruitment process and administrators can monitor the platform.

1.6 Terms, Acronyms, and Abbreviations

The following terms, acronyms, and abbreviations are used throughout this report:

Table 1.1: Terms, Acronyms, and Abbreviations Used in CareerConnectBD

Term/Acronym	Description
CareerConnectBD	The web-based career and recruitment management platform developed as this project.
Admin	Administrator responsible for managing and monitoring major platform resources and activities.
Job Seeker	A user who creates a career profile, manages resumes, searches for jobs, and submits job applications.

Term/Acronym	Description
Employer	A user who manages company information, publishes job opportunities, and manages recruitment activities.
MVC	Model-View-Controller, the architectural pattern used by the Laravel framework to organize application components.
RBAC	Role-Based Access Control, an access-control approach in which system permissions are determined according to user roles.
CRUD	Create, Read, Update, and Delete, representing fundamental data management operations.
ORM	Object-Relational Mapping, a technique used to interact with relational databases through application objects. Laravel Eloquent is used as the ORM in this project.
ATS	Applicant Tracking System, a term associated with systems and processes used to organize and evaluate recruitment applications and resume-related information.
UI	User Interface, the visual components through which users interact with the system.
UX	User Experience, referring to the overall experience of users while interacting with the application.
PDF	Portable Document Format, a document format used for resume-related preview or downloadable documents where supported by the system.
API	Application Programming Interface, a mechanism that allows software components or external services to communicate programmatically.
CSRF	Cross-Site Request Forgery, a type of web security attack against which Laravel provides request protection mechanisms.
DB	Database, referring primarily to the MySQL relational database used by CareerConnectBD.
AI	Artificial Intelligence. In this report, AI-related features refer to potential future enhancements; the current implementation focuses on role-based workflows, structured data management, and standard search and filtering capabilities.

Project Management Plan

2.1 Project Organization

CareerConnectBD is a final year software engineering project, created as a group effort. The project work was organized around the three main roles of a user (Job Seeker, Employer, Administrator) and the functional modules (Authentication and Authorization (RBAC), Job and Application Management, Resume Management, Recruitment Workflows, Notifications, Support Ticket Management, Feedback Management, and Administrative Oversight).

The team followed a collaborative development process with version control using Git and GitHub. The requirements and the tasks were monitored and tracked iteratively, the system was integrated and tested continuously to minimize the merge conflicts and to find the regressions at an early stage.

2.1.1 Individual Contribution

Table 2.1 summarizes the confirmed primary responsibilities of each team member during the development of CareerConnectBD.

Table 2.1: Individual Contribution to CareerConnectBD

Team Member	Primary Contribution
Md. Tareq Miah	Project planning and requirement analysis; Laravel 13 and PHP backend development; MySQL database design and database migrations (Eloquent ORM); authentication implementation and Laravel Breeze integration; Role-Based Access Control (RBAC); integration of the Job Seeker module, Employer module, job management and job application workflow, recruitment workflow, notifications, support tickets, and feedback modules; system testing and debugging; Git/GitHub integration and version-control coordination; technical documentation and final project report coordination.
Asif Mahmud Khan	Vue.js frontend development; Bootstrap 5 responsive user interface development; frontend component development; employer-related and recruitment-related user interfaces; frontend-backend integration; UI testing and frontend debugging; usability improvement and team integration support.

Team Member	Primary Contribution
Sunia Akther	Vue.js frontend development; Job Seeker user interfaces and general user-facing interfaces; responsive UI/UX implementation; frontend component integration; support for form and interface validation; UI testing and frontend debugging; documentation support and presentation preparation support.

The team members had specific main roles but the development of CareerConnectBD was a collective effort. Coordinating the integration of the system, testing, debugging, resolving issues, and preparing the final project involved all three members, ensuring the seamless harmony between the backend, frontend, database, and user-interface elements of the system.

2.2 Process Model Used

This was done as an iterative and incremental development process. The work was broken down into small cycles where attention was devoted to a group of requirements (authentication and RBAC; job posting and applications, resume services, notifications, administrative operations and more). This approach enabled the requirements to be continually refined and it reduced risk because the software artefacts were created that were able to be used to experiment with.

2.2.1 Rationale for choosing lifecycle model

Many different modules and role-specific workflows had to be integrated into the platform so the iterative lifecycle model was selected. It was possible to incrementally deliver the new features such as resume builder, recruitment workflows, notification, support tickets and feedback etc. and the end to end validation (login, role based dashboards, job application flow etc.) later on. Additionally, the regular integration allowed for the changes in one module not to inadvertently break other modules.

2.3 Feasibility Study

2.3.1 Technical Feasibility

CareerConnectBD was technically viable because existing web technologies and frameworks were used to develop it, as part of an academic project. Laravel 13 and PHP 8.4 offered a secure and stable backend platform that can be structured using MVC, and included ORM support. Relational data storage was done with MySQL, and Bootstrap 5, Laravel Blade and Vue.js/JavaScript (when it was around) were used for the UI. Local development, testing and deployment preparation were facilitated by open-source tooling (Composer, node.js/NPM, git).

2.3.2 Economic Feasibility

The project had an economic impact in an academic environment, as it was based on open-source development tools and frameworks, and can be run on a standard personal

computer. The two main costs are developer time and (optionally) deployment infrastructure. The project can thus be implemented without large direct financial investments.

2.3.3 Operational Feasibility

The system is feasible to operate as it uses repetitive workflows on a job portal for Job Seekers and Employers, and offers an administrative interface to oversee the system. Role based access control can enable presentation features based on the role of the user, for better usability and security. The notification mechanism and support/feedback functionality (where available) enhance communication along the platform.

2.3.4 Schedule Feasibility

Because the project was broken up into modules and performed incrementally, it was schedule feasible. Based on the logic of the development task, user roles and the complexity of the flow, the development was grouped so that the development could be done in parallel by each team member and the dependency bottlenecks were minimized.

2.4 Risk Analysis

Key project risks and mitigations are summarized below.

- **Scope expansion risk:** Requirements may expand beyond the available academic schedule. *Mitigation:* Prioritize core workflows (authentication, RBAC, job posting, job browsing, applications, basic administration) before enhancements.
- **Integration risk:** Modules may conflict when combined (routes, authorization checks, database relationships). *Mitigation:* Continuous integration through frequent Git commits, code reviews, and regular end-to-end testing.
- **Security risk:** Improper authorization may expose protected resources. *Mitigation:* Enforce RBAC using middleware, policies, validation, and ownership checks; test negative and unauthorized scenarios.
- **Data consistency risk:** Complex relationships (jobs, applications, interviews, notifications) may lead to inconsistent data. *Mitigation:* Use relational constraints and Laravel validation; verify workflows using system-level test cases.
- **Deployment configuration risk:** Environment differences may cause runtime failures. *Mitigation:* Document installation and configuration steps; keep environment-specific settings in `.env`.

2.5 Constraints to project implementation

The project was implemented under common academic constraints.

- **Schedule constraints:** Limited academic timeline required prioritization of core features.
- **Budget constraints:** Development relied on open-source tools and local environments; production hosting was treated as optional.

- **Software constraints:** Compatibility constraints included Laravel and PHP version requirements and Node.js tooling for frontend builds.
- **Hardware constraints:** Development and testing were performed on personal computers with limited resources compared to production servers.

2.6 Hardware and Software Resource (Tools/Language) Requirements

Hardware (minimum for development):

- Personal computer with multi-core CPU
- 8 GB RAM (recommended 16 GB for smoother builds)
- Sufficient storage for source code, dependencies, and database (recommended 10+ GB free space)

Software:

- PHP 8.4, Composer, Laravel 13
- MySQL server
- Node.js and NPM (for Vite-based frontend development)
- Git and a GitHub repository for collaboration
- Visual Studio Code (or an equivalent IDE)

2.7 Project Timeline and Schedule (Gantt Chart)

CareerConnectBD was planned and implemented within 6 project months (M1 - M6). The sequence included requirement analysis, planning, feasibility analysis, requirement specification, system design, database design, setting up a development environment, designing authentication and authorization, developing the core module, developing the frontend and integrating it with the core module, preparing for the recruitment process, designing supporting modules, testing and debugging, documentation, and preparation for presentation and defense. Several activities overlapped as the frontend, backend development, integration, security and testing was done in an incremental manner throughout the project lifecycle.

Table 2.2: Six-Month Development Timeline (Gantt Chart) of CareerConnectBD

Project Activity	M1	M2	M3	M4	M5	M6
Requirement Analysis & Project Planning	✓					
Feasibility Study & System Requirement Specification	✓					
System Design, DFD, Use Case & Database Planning	✓	✓				
Laravel Environment & Database Setup	✓	✓				

Project Activity	M1	M2	M3	M4	M5	M6
Authentication & Laravel Breeze Integration		✓				
Role-Based Access Control (RBAC)		✓				
Job Seeker Profile Management		✓	✓			
Resume, Education, Experience & Skills Management			✓			
Employer Profile & Company Management				✓		
Job Posting & Job Management		✓	✓			
Job Search & Job Application Workflow				✓		
Employer Application Management				✓		
Interview Invitation & Response Workflow				✓	✓	
Notification System					✓	
Support Ticket & Feedback Module					✓	
Vue.js / Bootstrap Frontend Integration	✓	✓	✓	✓		
UI/UX Improvement & Responsive Design			✓	✓		
Security, Validation & Authorization	✓	✓	✓	✓		
Integration & System Testing					✓	✓
Bug Fixing & Performance Improvement					✓	✓
Final Documentation & Project Report					✓	✓
Final Presentation & Defense Preparation						✓

2.8 Estimated Budget

Since the system was developed as an academic project using open-source tools, the budget is best presented as an estimated development budget rather than as verified expenditure. Typical cost categories include:

- Development hardware (personal computers)
- Internet connectivity
- Optional hosting/domain costs for deployment testing (if used)
- Developer time (academic effort)

2.9 Social/Cultural/Environmental impact of the project

CareerConnectBD can be a source of social impact by increasing access to information about opportunities, and helping employers and job seekers to recruit. The system supports and allows for structured career profile development and tracking applications, with some possible reduction in manual communication during the recruitment process.

The platform's cultural aspects facilitate regular hiring processes and the management of resumes and profiles. Role based views help achieve a 'user experience that matches the role and the organisation.

The project is low impact on the environment, primarily because of the use of computing resources. It's possible for digital recruitment workflows to mitigate paper-based recruitment and travel for initial screening. It is possible to deploy the system on any regular web infrastructure and can be used on a sustainable basis by its efficient database design, optimized builds of the front end and responsible settings of the servers.

Requirement Analysis and Specification

3.1 Requirement Analysis

CareerConnectBD is a career and recruitment management system that is web-based and is aiming for an integrated job seeker, employer and administrator environment. The system requirement analysis included the identification of the major roles of the users, the duties of each role, expected system behavior, data requirements and administrative operations.

There are three distinct user roles in the system: Job Seeker, Employer and Admin. Role based authorizations are used to provide a specific set of functionalities to each role.

Job seekers need facilities to register and log into accounts, to manage their profiles, to manage their education and experiences, to manage their skills, to build their resume, to search for jobs, save their jobs, apply for a job, track their application, participate in interviews, receive notifications, access support tickets and respond to them, and give feedback.

Employers need to have facilities for registering and authenticating their accounts, manage their company profiles, post and maintain their jobs, review resumes of candidates, manage applications status, review resumes prior to interview, invite candidates for interviews, manage employee notifications, and other employer-specific activities.

Administrators need centralized platforms to check and manage the platform. These include dashboard statistics, user management, employer and company management, job and application monitoring, reports, system settings, support ticket management, feedback management and more.

Analysis of system requirements focussed on usability, security, maintainability, data integrity, role based access control and dependable interaction between three major user roles.

3.2 Functional Requirements

The functional requirements outline what operations and services CareerConnectBD users are provided. The following descriptions of the major functional requirements outline the

purpose of the components.

User Authentication and Authorization

- The system should enable a job seeker to register for an account.
- The system shall enable employers to sign up for an employer account.
- The system should enable the secure login and logout of registered users.
- The system should implement role-based access control for Admin, Employer and Job Seeker users.
- The system shall redirect authenticated users to their appropriate area based on their role.
- The system will not allow unauthorized users to access protected resources.

Job Seeker Management

- Job seekers should be able to make and edit their own profile.
- Job seekers shall be able to handle the information relating to education.
- Job seekers will be able to handle information about professional experiences.
- Job seekers shall be able to manage their skills.
- Job seekers should be able to upload/resume and manage resumes.
- Job seekers must be able to use the resume builder feature of the system to create a resume.
- Job seekers will be able to view and download the resume files they support.

Job Search and Saved Jobs

- Users should be able to view a selection of job opportunities.
- Each user should have access to detailed information regarding a job.
- Users looking for jobs shall be able to save the records for later viewing.
- Job seekers will be able to see and edit jobs that they save.

Job Application Management

- All registered users seeking employment should be able to apply for jobs available.
- Where applicable, the system should not allow invalid job applications or duplicate job applications.
- Job seekers will be able to see applications they have submitted.
- Job seekers shall have the opportunity to keep an eye on their applications.
- Employers can view the applications applied to their company's jobs.

- Employers should have the ability to change the status of an application in line with the recruitment process.

Employer and Company Management

- Employers will be able to develop and maintain company information.
- Employers will have access to an employer specific dashboard.
- Employers should have the ability to make job postings.
- Employers should have control over making changes to their ads and over their ads.
- Employers should be able to control the deletion of job records if the job-management workflow implemented allows.
- Employers may check applicants' application for their positions.
- Employers shall have the ability to view or print out candidate resumes (where legal by system).

Interview and Recruitment Management

- Employers will be able to control the activities carried out in the interview that were facilitated by the system.
- Employers shall be able to send interview invitation to appropriate candidates.
- Job seekers are entitled to information regarding interview.
- Candidates should be able to accept invitations for interviews when they are invited and the workflow has been implemented.
- Notifications shall be created for relevant recruitment actions for the proper users.

Notification Management

- The system shall be notifiable of user activities relevant to the database.
- Employers will be notified when job seekers apply for their jobs.
- The job seekers will be notified of changes to application statuses if applicable.
- Job seekers will be notified of interview invitations that are relevant.
- Notices will be sent to employers for appropriate candidate interview responses.
- Administrators will be notified of relevant support requests and feedback sent.
- Users should be able to see their notifications and control when they've read them.
- The core database notification delivery may be used independently of realtime notification broadcasting.

Admin Management

- Administrators can be given access to a secured admin dashboard.
- Administrators can see system-wide statistics and summaries.
- Administrators will be able to control users of the platform.
- Administrators shall have the authority to control the status of users and/or to conduct appropriate administrative functions related to their role.
- Administrators will have the ability to track employers and companies.
- Administrators will have the ability to view jobs and applications.
- Administrators will have access to system reports and platform metrics.
- Administrators shall have access to and be able to control certain system settings.
- Administrators will be able to handle tickets of support and reply to users.
- Administrators will have access to user feedback.

Support Ticket and Feedback Management

- Authorized users will be able to submit support tickets.
- Administrators will have access to support tickets submitted.
- Administrators should have the ability to respond to support tickets.
- Appropriate users will be informed of appropriate ticket activity.
- The user(s) should be able to provide feedback via the feedback mechanism provided.
- Administrators will be able to see submitted feedback.

3.3 Non-Functional Requirements

All the non-functional requirements outline the quality attributes and operational characteristics that are expected of CareerConnectBD.

Security

The system has to offer safe authentication and role-based authorizations. Protected routes and resources are to be used by authorized personnel only. The application will implement the security features offered by the Laravel framework, such as password hashing, validation of incoming requests, CSRF protection, authentication middleware, and role-based access control.

Usability

The user interface shall be intuitive and have functionality relevant to the user's role. The interface structure of the application shall be consistent with the purpose of the

application to enable job seekers, employers and administrators to easily access desired features.

Performance

The application will process usual user requests efficiently and fetch any required information from the database without unnecessary delay in the anticipated usage of the project. The structure of the database and the logic of the applications should be designed in a way that allows for efficient retrieval and processing of data.

Reliability

Core business operations should be reliable when used normally in application. Important information such as user profiles, jobs, applications, and notifications shall be stored persistently in the database. Core database-backed notification operations shall not be prevented by the fact that the realtime notification service is not available.

Maintainability

The application should be built using a MVC architecture, and it should not mix the presentation and application logic together with the data access concerns. Controllers, Models, Middleware, Services, Requests, Policies and Blade Views shall be structured to enable future maintenance and extension.

Scalability

Architecture should accommodate future user growth, job postings, applications, companies and other resources of the platform. Laravel's modular structure allows potential extension of features as needed.

Data Integrity

The system will ensure proper relationships between users, roles, job seeker profiles, companies, jobs, applications, resumes, and other related entities. To minimize invalid and inconsistent data, use validation rules and database relationships.

Compatibility

The application shall be usable using modern web browsers. If implemented, the responsive user interface created with Bootstrap will be compatible with various common screen sizes.

3.4 Data Flow Diagram (DFD)

Data Flow Diagrams are used to illustrate how information moves between external entities, system processes, and data stores within CareerConnectBD. The diagrams represent the interactions of Job Seekers, Employers, and Administrators with the major processes of the platform.

3.4.1 Context Diagram (Level 0 DFD)

The Context Diagram depicts CareerConnectBD as a single system, and shows how it communicates with the three main external systems: Job Seeker, Employer and Admin. Figure 3.1 shows the context-level data flow of the platform.

The Job Seeker inputs his/her registration information, profile, resume, information regarding job applications, answers to questions during interview, support requests, and feedback into the system. The system, in turn, gives the information of the job, application status information, interview information, notification and other career related services.

The Employer will provide details regarding the company, posting details, recruitment decision, the application status and details regarding the interview. The system offers candidate applications, resume details, recruitment information, and pertinent notifications.

An Admin communicates with the system by interacting with users, reviewing reports, managing system resources, support tickets and feedback, and setting up supported system settings. The system is used for administrative information, statistics, reports and data for management.

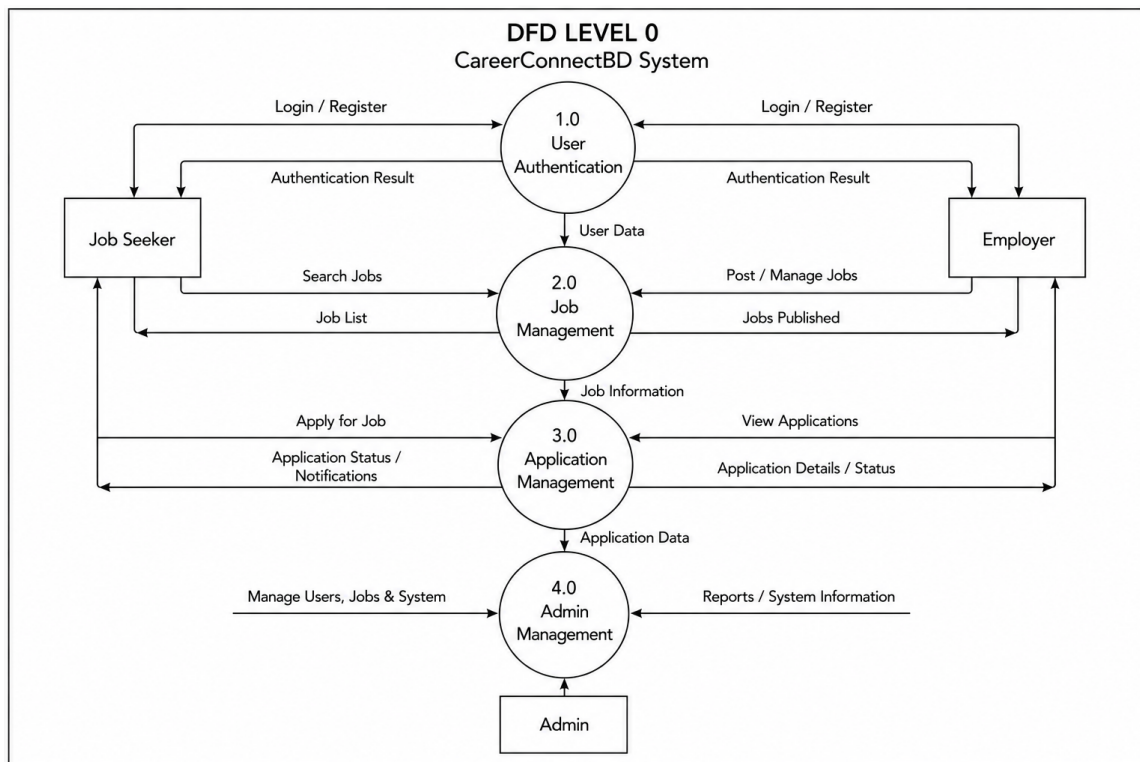


Figure 3.1: Context Diagram (Level 0 DFD) of CareerConnectBD

3.4.2 Level 1 DFD

The Level 1 DFD breaks down CareerConnectBD into its key functional processes. These processes consist of User and Authentication Management, Job Seeker Profile and Resume Management, Company and Job Management, Job Application Management, Interview

and Recruitment Management, Notification Management, Support and Feedback Management and Administrative Management. This decomposition is shown in Figure 3.2.

Key Data Stores include user and role, job seeker profile, company, job, application, resume, interview, notification, support ticket, and feedback, and user data.

The Job Seeker's interactions are predominantly with the profile and resumes management, job searching, and job application processing, as well as interview activity, notification and support for the job search process. The Employer will be responsible for handling primarily the company management, job management, application processing, interview management and notifications. The Admin is communicating with administrative management, reporting, user management, support tickets, feedback and system monitoring.

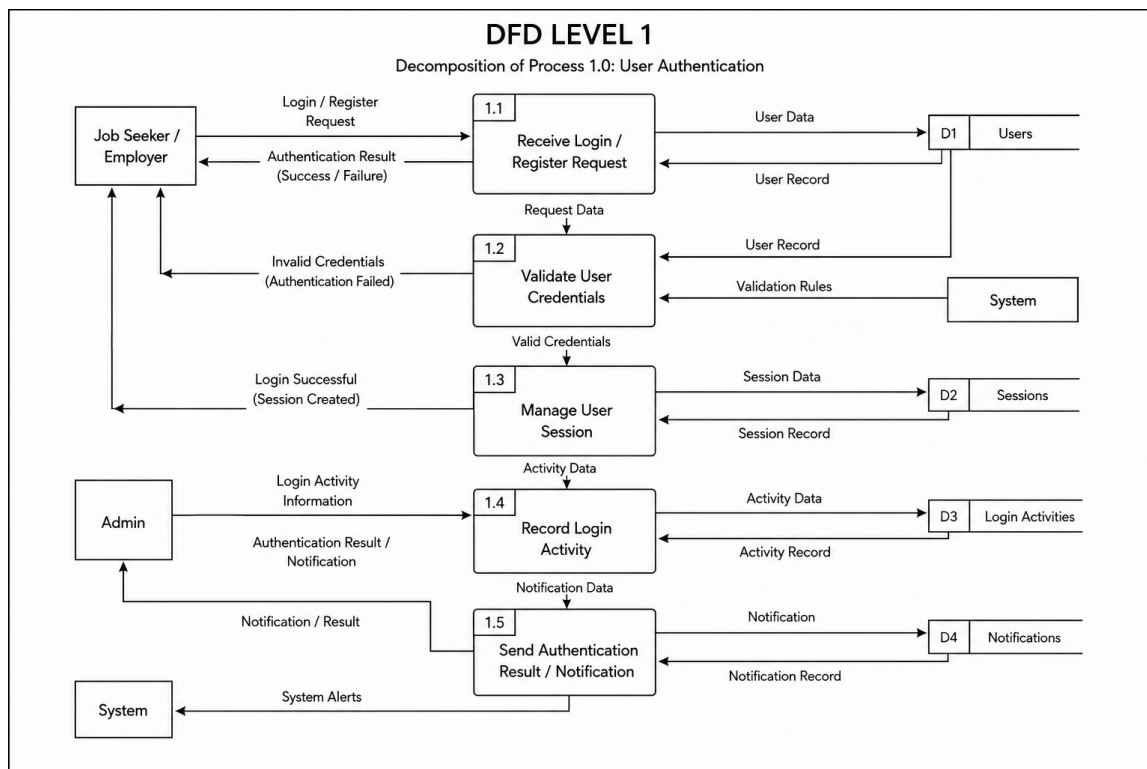


Figure 3.2: Level 1 Data Flow Diagram of CareerConnectBD

3.5 Use Case Diagram with Graphical and Textual Description

The Use Case Diagram shows the significant interactions between the system actors and CareerConnectBD. The key players include Guest User, Job Seeker, Employer and Administrator.

Job Seeker use cases include registering, logging in, managing their profile (education, experience, skills), managing their resume, browsing for jobs and saving jobs, job application, application tracking, interview related activities, notifications, submission of support tickets and giving feedback.

Employer related use cases are Register, Login, Company profile management, company job posting and management, job application review and resume review, application status update, interview management, notification view, etc. and Employer related recruitment activities.

Administrator Use Cases are: Dashboard Monitoring, User Monitoring, Job Management, Change Employer/Company Monitoring, Application Management, Viewing Reports, System Settings Management, Support Ticket Management, Reviewing Feedback, and Notification Activities.

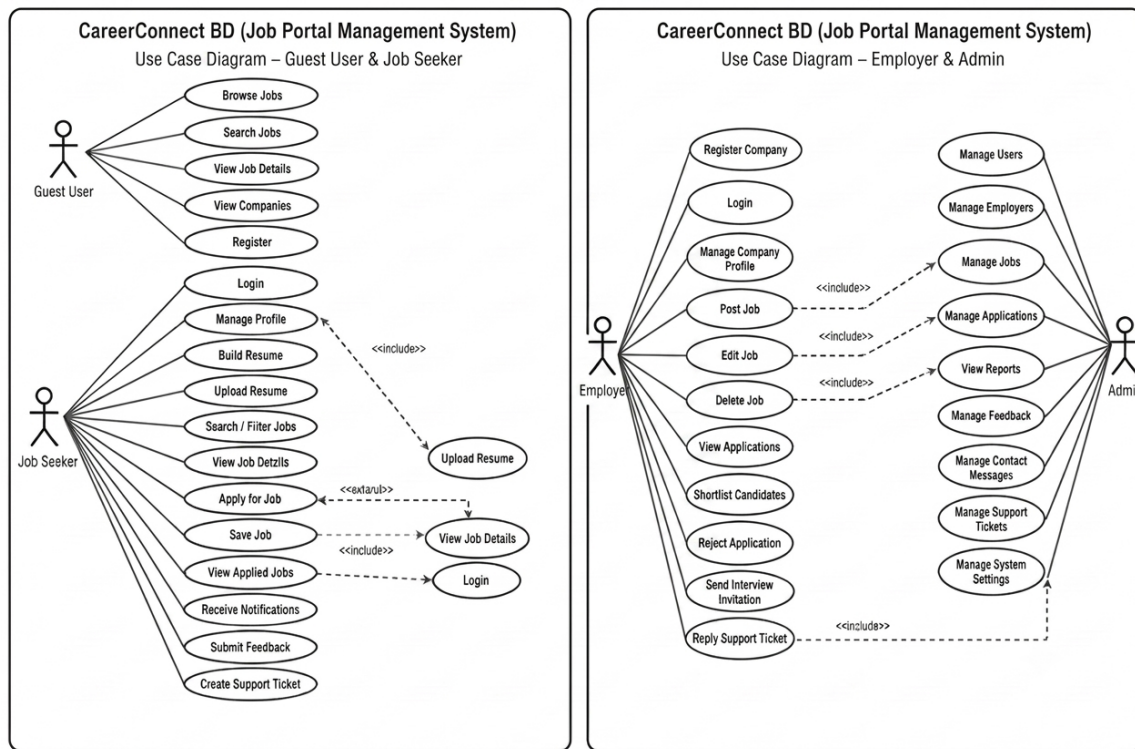


Figure 3.3: Use Case Diagram of CareerConnectBD

Graphical use case diagram is a high level description of actors involved with the system and their interactions with it. The descriptions below are brief and textual use case representations of the main workflows already being supported by CareerConnectBD.

3.5.1 Textual Use Case Descriptions

Table 3.1: Textual Description of the User Registration and Login Use Case

Use Case Name	User Registration and Login
Primary Actor(s)	Guest User, Job Seeker, Employer, Administrator
Preconditions	The user is not authenticated; the registration and login pages are accessible.

Main Flow	1. Guest User opens the registration page and selects the intended role (Job Seeker or Employer). 2. User submits the required registration information. 3. System validates input and creates the account. 4. User logs in using credentials. 5. System authenticates the user and redirects to the role-specific dashboard or landing page.
Alternative / Exception Flow	• Duplicate email/username or invalid input: system shows validation errors and registration is not completed. • Invalid credentials: system denies login and prompts the user to retry.
Postconditions	A valid user account exists (for registration) and an authenticated session is established (for login).

Table 3.2: Textual Description of the Manage Job Seeker Profile Use Case

Use Case Name	Manage Job Seeker Profile
Primary Actor(s)	Job Seeker
Preconditions	Job Seeker is authenticated and has access to the profile module.
Main Flow	1. Job Seeker navigates to the profile section. 2. Job Seeker views and updates personal and career-related information (including education, experience, and skills where applicable). 3. Job Seeker submits the updated information. 4. System validates and stores the changes.
Alternative / Exception Flow	• Missing/invalid required fields: system shows validation messages and does not save changes. • Session expired: system requests the user to log in again.
Postconditions	The Job Seeker profile information is updated and persisted in the system.

Table 3.3: Textual Description of the Manage Resume Use Case

Use Case Name	Manage Resume
Primary Actor(s)	Job Seeker
Preconditions	Job Seeker is authenticated and has access to the resume management module.

Main Flow	1. Job Seeker opens the resume management section. 2. Job Seeker creates/updates resume content (and/or uploads an updated resume where supported). 3. Job Seeker saves the resume. 4. System stores the resume and makes it available for job applications.
Alternative / Exception Flow	• Unsupported or invalid resume file/content: system displays an error and requests correction. • Save failure (e.g., incomplete data): system does not update stored resume and prompts retry.
Postconditions	Resume information is stored/updated and can be referenced during job applications.

Table 3.4: Textual Description of the Search and View Jobs Use Case

Use Case Name	Search and View Jobs
Primary Actor(s)	Guest User, Job Seeker
Preconditions	The job listing and job details pages are accessible.
Main Flow	1. User navigates to the job listings page. 2. User searches and/or filters jobs using available criteria. 3. System returns a list of matching jobs. 4. User opens a job to view job details.
Alternative / Exception Flow	• No matching jobs found: system displays an empty result message and allows refining the search. • Job is unavailable/closed: system restricts actions accordingly (e.g., applying).
Postconditions	Matching jobs and/or selected job details are displayed to the user.

Table 3.5: Textual Description of the Apply for Job Use Case

Use Case Name	Apply for Job
Primary Actor(s)	Job Seeker
Preconditions	Job Seeker is authenticated; a target job is available; required application information (e.g., resume) is available.

Main Flow	1. Job Seeker opens a job details page. 2. Job Seeker selects the apply option and confirms required application inputs. 3. System validates the application and stores it under the Job Seeker account. 4. System notifies relevant parties according to the application workflow.
Alternative / Exception Flow	• Missing resume or required details: system prompts the user to complete resume/profile information before applying. • Duplicate application detected: system prevents re-submission for the same job.
Postconditions	A job application record is created and becomes visible in application tracking for both Job Seeker and Employer.

Table 3.6: Textual Description of the Track Job Applications Use Case

Use Case Name	Track Job Applications
Primary Actor(s)	Job Seeker
Preconditions	Job Seeker is authenticated and has submitted at least one job application.
Main Flow	1. Job Seeker navigates to the application tracking (My Applications) module. 2. System displays the list of submitted applications with current status. 3. Job Seeker opens an application to view details and any status updates (including interview-related updates where applicable).
Alternative / Exception Flow	• No applications found: system displays an informative message.
Postconditions	The Job Seeker has visibility into the current state of submitted applications.

Table 3.7: Textual Description of the Manage Employer/Company Profile Use Case

Use Case Name	Manage Employer/Company Profile
Primary Actor(s)	Employer
Preconditions	Employer is authenticated and has access to company profile management.

Main Flow	1. Employer navigates to the company profile section. 2. Employer views and updates company information. 3. Employer saves the changes. 4. System validates and stores updated company profile data.
Alternative / Exception Flow	• Invalid/missing required fields: system shows validation errors and does not save. • Session expired: system requests re-authentication.
Postconditions	The company profile information is updated and persisted in the system.

Table 3.8: Textual Description of the Post and Manage Jobs Use Case

Use Case Name	Post and Manage Jobs
Primary Actor(s)	Employer
Preconditions	Employer is authenticated; company profile information is available; employer has access to the job management module.
Main Flow	1. Employer navigates to the job management section. 2. Employer creates a new job post by providing required job details. 3. System validates and publishes/saves the job post. 4. Employer edits, updates, or closes job posts as needed.
Alternative / Exception Flow	• Missing or invalid job information: system prevents publishing and requests corrections. • Job post update fails: system retains the previous saved state and prompts retry.
Postconditions	Job posts are created/updated/closed and become available for job seeker browsing according to their status.

Table 3.9: Textual Description of the Manage Job Applicants Use Case

Use Case Name	Manage Job Applicants
Primary Actor(s)	Employer
Preconditions	Employer is authenticated; at least one job post exists; applications are available for review.

Main Flow	1. Employer selects a job post from the employer dashboard/job management module. 2. System displays the list of applicants and their submitted information (including resumes). 3. Employer reviews applicant details. 4. Employer updates the application status according to the recruitment workflow. 5. System stores the updated status and triggers relevant notifications.
Alternative / Exception Flow	• No applicants found for a job post: system displays an informative message. • Invalid status update or access issue: system rejects the change and prompts corrective action.
Postconditions	Application statuses are updated and visible to the Job Seeker through application tracking.

Table 3.10: Textual Description of the Interview Invitation and Response Use Case

Use Case Name	Interview Invitation and Response
Primary Actor(s)	Employer, Job Seeker
Preconditions	Employer is authenticated; a job application exists and is eligible for interview action.
Main Flow	1. Employer selects a candidate application and initiates an interview-related action. 2. System creates an interview invitation/entry and notifies the Job Seeker. 3. Job Seeker views the interview invitation from the relevant module/notifications. 4. Job Seeker responds (e.g., accept/decline) where supported by the workflow. 5. System records the response and notifies the Employer.
Alternative / Exception Flow	• Job Seeker does not respond within the expected time-frame: invitation remains pending until updated. • Invalid scheduling or missing details: system prompts the Employer to correct the invitation information.
Postconditions	Interview invitation status and Job Seeker response are recorded and reflected in the recruitment workflow.

Table 3.11: Textual Description of the Notification Management Use Case

Use Case Name	Notification Management
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Primary Actor(s)	Job Seeker, Employer, Administrator
Preconditions	User is authenticated; notification events are generated by the system modules.
Main Flow	1. System generates notifications for relevant events (e.g., application status updates, interview activities, support ticket updates). 2. User opens the notifications section. 3. System displays notification list and details. 4. User reads and optionally marks notifications as read.
Alternative / Exception Flow	• No notifications available: system shows an empty state. • Temporary delivery issue: notification is still persisted and becomes visible when the user accesses the platform.
Postconditions	Notifications are presented to the user and read/unread status is updated where supported.

Table 3.12: Textual Description of the Support Ticket and Feedback Management Use Case

Use Case Name	Support Ticket and Feedback Management
Primary Actor(s)	Job Seeker, Employer, Administrator
Preconditions	User is authenticated; support/feedback module is accessible.
Main Flow	1. Job Seeker or Employer opens the support/feedback section. 2. User submits a support ticket and/or feedback with relevant details. 3. System stores the submission and notifies Administrator as applicable. 4. Administrator reviews submitted tickets/feedback and performs management actions (e.g., respond, update status, resolve).
Alternative / Exception Flow	• Missing required details: system requests the user to provide sufficient information. • Unauthorized access attempt: system denies the action.
Postconditions	Support tickets/feedback are stored, tracked, and updated according to administrative actions.

Table 3.13: Textual Description of the Administrative Management Use Case

Use Case Name	Administrative Management
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Primary Actor(s)	Administrator
Preconditions	Administrator is authenticated and authorized to access the administrative dashboard.
Main Flow	1. Administrator logs in and opens the administrative dashboard. 2. Administrator performs management activities such as user management, employer/company monitoring, job and application management, report viewing, and system settings management. 3. Administrator reviews and manages support tickets and feedback. 4. System applies changes and updates the relevant modules and notifications.
Alternative / Exception Flow	• Unauthorized user attempts administrative action: system denies access. • Invalid management input: system rejects the change and prompts correction.
Postconditions	Administrative updates are applied to the platform data and reflected across relevant system modules.

3.6 Activity Diagram

The Activity Diagram shows the overall flow of the CareerConnectBD system from user authentication to the system activities based on the user's roles. Summarizing this flow, we can see the end-to-end flow in Figure 3.4.

The starting point is that a user logs into the platform. First, a user logs into the platform. An unauthenticated user can view information that is publicly available, or register or log in. If authentication is successful, the system determines the specific role assigned to the user and grants access to the proper functionality.

A Job Seeker can update profile and career information, handle resumes, job search, save jobs, apply for job(s), track application status, engage in interview workflows, and be notified.

An Employer can store company data, post and manage job vacancies, review applications, update the status of applications managed by the Employer, initiate interview-related activities, and get notified of the application received on their recruitment.

The administrative dashboard is only accessible by an Admin, and they can use it to manage users, monitor employers and companies, create and monitor jobs and applications, view reports, do system settings, handle support tickets and review feedback.

The workflow proceeds until the user finishes the activity or logs out of the system, based on the operation(s) selected by the user.

Activity Diagram – CareerConnectBD (Job Portal Management System)

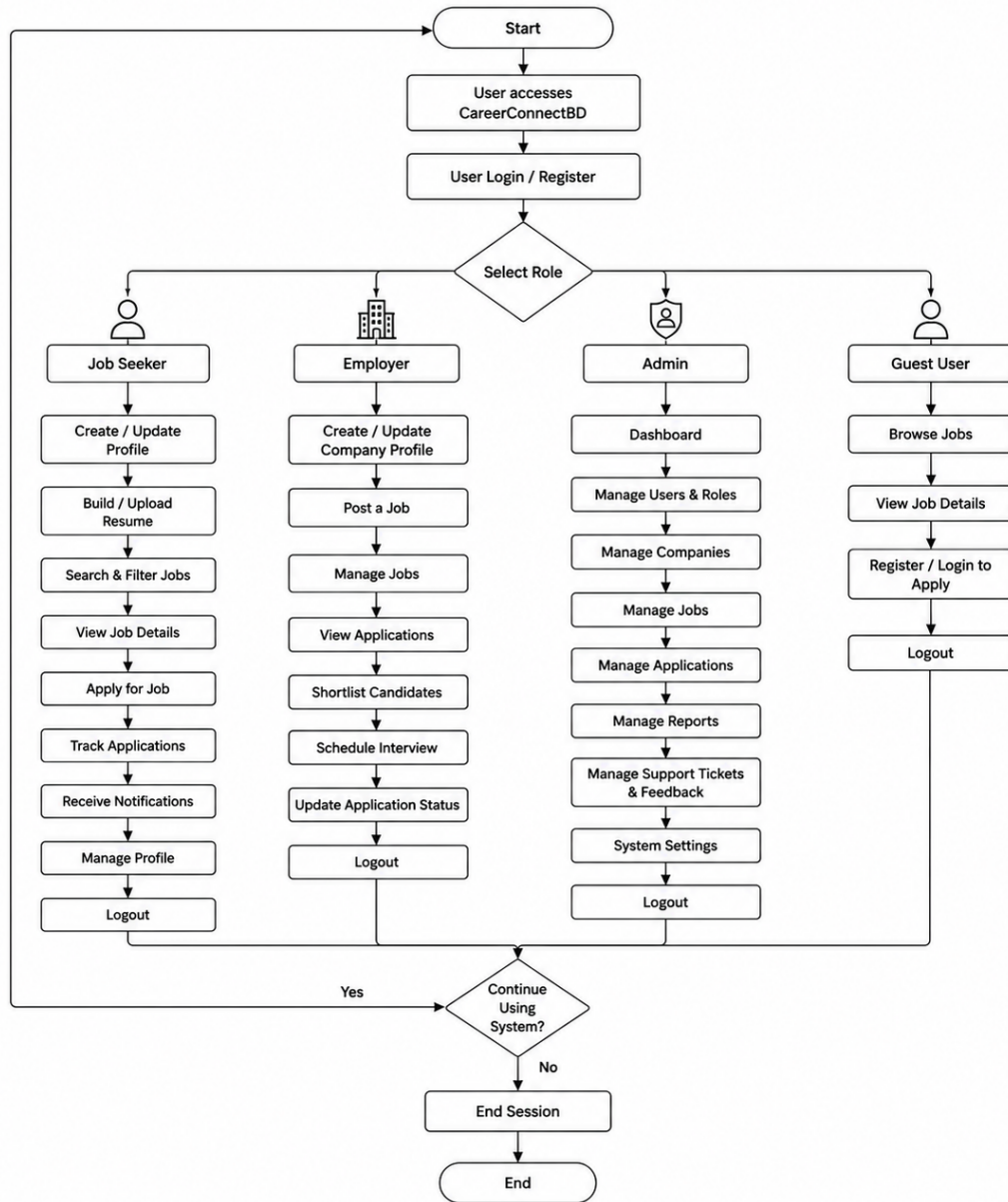


Figure 3.4: Activity Diagram of CareerConnectBD

System Architecture

4.1 Architectural Model/Style Used

CareerConnectBD is implemented as a web-based career and recruitment management platform (CRMP) with Model–View–Controller (MVC) pattern designed by the Laravel framework [1]. MVC is a three-layer application design which provides an abstraction of the Model, View and Controller concerns. This separation leads to better maintainability, modularity, scalability, and code organization.

The **Model** layer is the layer that represents the Application’s data and Business entities. It is integrated with the MySQL relational database system, through the use of the Laravel Eloquent ORM (Object Relational Mapping). The main components of the system are Users, Roles, Job Seeker Profiles, Employers, Companies, Jobs, Job Applications, Education Records, Experience, Skills, Resumes, Resume Builders, Interview related records, Notifications, support ticket, feedback, etc.

The **View** layer is exposed to the users and offers interactive user interfaces. Vue.js is used for the frontend to enable interaction and create components [2] and Bootstrap 5 is used for the responsive styling, layout design and reusable user interface components. Laravel Blade is also used for server-rendered views, layouts and integration with the Laravel backend (if applicable) [1]. For JavaScript, Vue.js and CSS assets, vite is used as the frontend development and build tool. The interfaces and dashboards are divided according to user roles of Job Seeker, Employer and Admin.

The **Controller** layer receives HTTP requests, checks user actions, coordinates user actions in the application, communicates with models and service classes, and provides responses or views. The major system modules that are put in place via controllers are authentication, profile administration, job administration, job applications, employer operations and administration, notifications and support and feedback.

CareerConnectBD also leverages a service-layer approach for selected business processes in addition to the MVC pattern. Service classes are used to separate complex business logic from controllers to make the controllers simpler for handling requests and the application easier to maintain and test.

In addition, the system uses role based access control (RBAC). User permission is granted based on user roles for authenticated users. There are 3 roles: Admin, Employer, and Job Seeker. Restriction of access to protected resources and operations with middleware, policies, form request validation, etc. are taken care of using the authorization mechanisms

provided in Laravel [1].

In summary, the architecture of CareerConnectBD is a layered MVC based architecture consisting of the following layers: Presentation Layer, Application/Controller Layer, Service/Business Logic Layer, Data Access Layer, MySQL Database Layer. The presentation layer is implemented using Vue.js, Bootstrap 5 and (if needed) Laravel Blade, while the application logic for the backend is provided by Laravel (PHP 8.4), and persistent relational data is stored in MySQL. The presentation layer is done with Vue.js, Bootstrap 5, and (if necessary) Laravel Blade, the application logic in the backend with Laravel (PHP 8.4) and persistent relational data stored in MySQL [3].

4.2 System Architecture Diagram

The CareerConnectBD system architecture diagram illustrates the relationship between users, the presentation layer, the application layer, the business logic/service layer, the data access layer, the database, and supporting systems. This is presented at a high level in Figure 4.1.

The three key players of the system are:

- Job Seeker
- Employer
- Admin

CareerConnectBD is accessed via a modern web browser. Vue.js provides interactive, component-based frontend functionality [2], Bootstrap 5 provides responsive styling and UI components [4], and (where applicable) Laravel Blade is used for server-rendered views and layouts [1]. Vite is used to build and serve frontend assets [5].

Laravel routes receive user requests and process them through middleware, authentication (Laravel Breeze) [6], and role-based authorization. Authorized requests are routed to the appropriate controllers. Controllers coordinate application operations and interact with service classes and Eloquent models to enforce business rules and perform database operations [7].

The service layer encapsulates selected business operations and helps keep complex application logic separate from HTTP request handling. Laravel Eloquent ORM provides the data access mechanism for application models and their relationships over the MySQL relational database [7, 8].

MySQL stores persistent data for the application [8]. Database notifications, file storage, resume management, interview workflows, support tickets, feedback management, administrative management, and reporting (as applicable) are included in the overall system architecture [1].

4.3 Technology Stack (Software and Hardware)

CareerConnectBD was developed using a modern web application technology stack. The backend is implemented using Laravel [1] and PHP [3], while Vue.js is used for inter-

System Architecture Diagram of CareerConnectBD

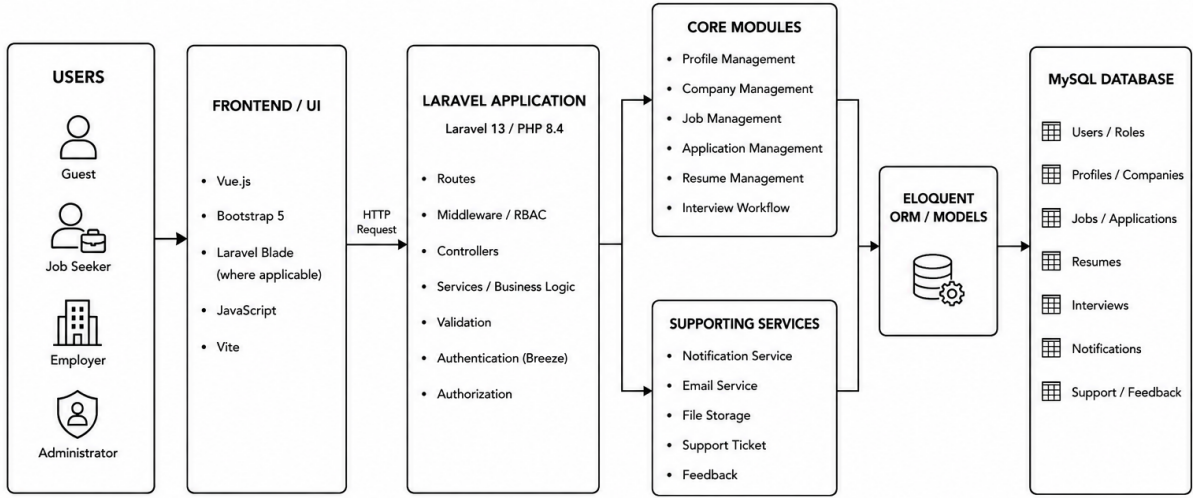


Figure 4.1: System Architecture Diagram of CareerConnectBD

active frontend development [2]. Bootstrap 5 provides responsive user interface styling [4], Laravel Blade supports server-rendered views where applicable [1], Vite is used for frontend asset bundling [5], and MySQL is used for persistent relational data storage [8]. The major software and hardware technologies used in the development and execution of the system are presented in Table 4.1.

Table 4.1: Technology Stack of CareerConnectBD

Category	Technology	Purpose
Backend Framework	Laravel 13	Used as the primary server-side web application framework and provides MVC architecture, routing, middleware, validation, authentication integration, ORM integration, and other application services.
Programming Language	PHP 8.4	Used for implementing backend application logic, controllers, models, services, validation, authentication, authorization, and server-side operations.
Frontend Framework	Vue.js	Used for developing interactive and component-based frontend functionality and dynamic user interfaces for the CareerConnectBD platform.
CSS/UI Framework	Bootstrap 5	Used for responsive styling, layout design, and reusable user interface components across public pages and role-based dashboards.

Category	Technology	Purpose
Template Engine	Laravel Blade	Used for server-rendered Laravel views, reusable layouts, and integration with frontend components where applicable.
Frontend Language	JavaScript	Used as the primary client-side programming language and for implementing interactive frontend functionality with Vue.js.
Frontend Build Tool	Vite	Used for frontend development and for building JavaScript, Vue.js, and CSS assets.
Database	MySQL	Used as the relational database management system for persistent application data.
ORM	Laravel Eloquent	Used for interacting with relational database tables through application models and relationships.
Authentication	Laravel Breeze	Used as the authentication foundation for user registration, login, logout, password management, and authenticated sessions.
Authorization	RBAC, Middleware and Policies	Used to control access to system features according to Admin, Employer, and Job Seeker roles.
Notification System	Laravel Notifications	Used for database-based user notifications and notification-related workflows within the platform.
Version Control	Git	Used for source code version control and collaborative development workflow.
Repository Hosting	GitHub	Used for hosting the project repository and supporting team collaboration and version management.
Development Environment	Visual Studio Code	Used as the primary source code editor and development environment.
Web Browser	Modern Web Browser	Used by Job Seekers, Employers, and Admins to access and interact with the CareerConnectBD web application.
Development Hardware	Personal Computer	Used for application development, frontend and backend implementation, testing, database management, and local execution of the system.

4.4 Database Design (ER Diagram)

CareerConnectBD has been built to use a relational database system, which was designed using MySQL [8]. The database stores structured relationships among the different components of the system, such as between users, roles, job seeker information, employers, companies, jobs, applications, resumes, recruitment activities, notifications, support activities and more.

The database structure follows a relational approach, using tables for entities and relationships between entities through primary and foreign keys. The database structure is defined and managed via migrations, while models are defined and give an object-oriented interface to the application's data via Eloquent ORM [7].

The user is the key component of the system. A user is assigned to a role, that defines what access rights and responsibilities he has. The primary roles in CareerConnectBD

are Admin, Employer and Job Seeker.

Job Seekers arrange information on their education, skills and resumes, and their work experience. The employers are responsible for conducting recruitment processes and for keeping information related to their companies on hand, according to their authorization. Companies are associated to jobs and Job Seekers can find jobs that are open and apply for them.

The application and recruitment flow connects Job Seekers, job applications, jobs, employers and companies. Additional entities in the database provide further support for resume management, resume building, education, work history, skills, interview workflows, notifications, support tickets, feedback and administration.

Entity Relationship (ER) Diagram – This diagram illustrates the main entities in the database and their relationships in a graphical format. A figure of the ER diagram for implementing the system is presented in figure 4.2.

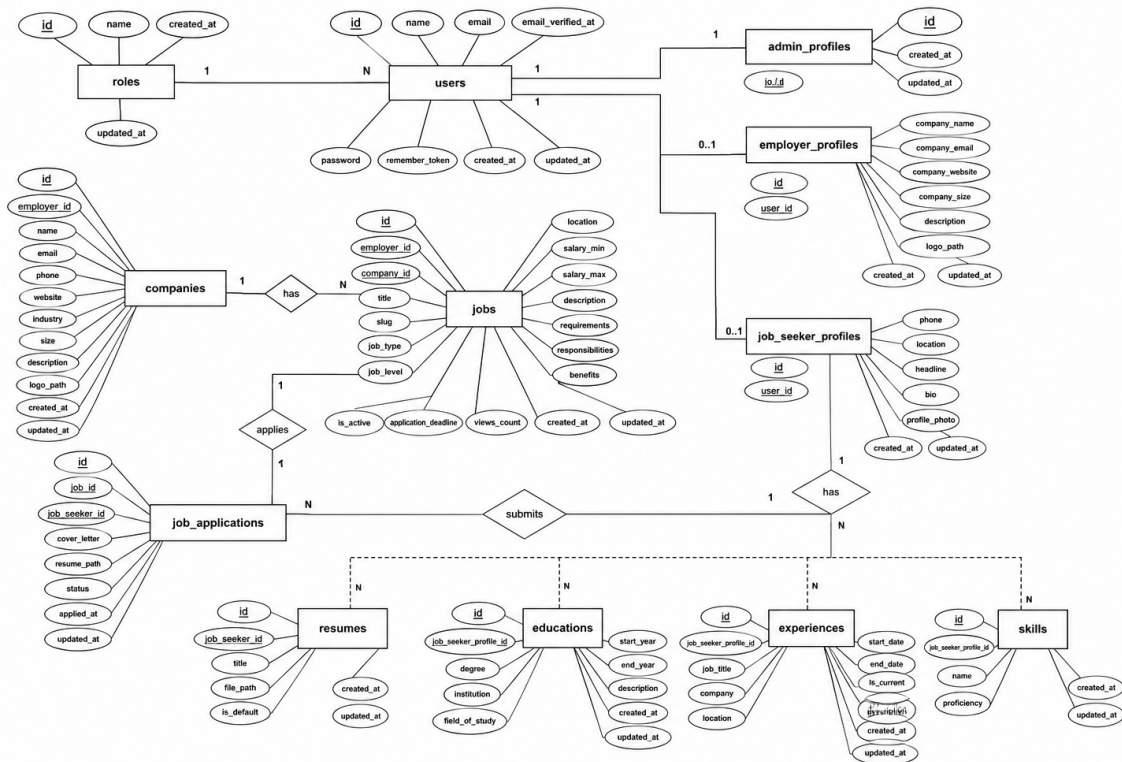


Figure 4.2: Entity Relationship (ER) Diagram of CareerConnectBD

4.5 Database Schema Design

The CareerConnectBD database schema has been designed to meet the business needs of the career and recruitment management platform. The schema contains logically related tables, and maintains relationships between tables through primary keys, foreign keys, and application-level relationships.

Laravel migrations are designed to define and control the structure of the database, and Laravel Eloquent ORM is used to create an interface for the application to interact with

the relational data.

The following are descriptions of the major logical groups of the database schema.

4.5.1 User and Role Management

The user and role management schema contains information about users and roles. The **users** table holds authentication and account information, and role attributes decide if the user is an Admin, Employer or Job Seeker.

This will enable role-based authentication and authorization system-wide. Role specific resources and operations are protected at the application layer using middleware, policies and other authorization mechanisms.

4.5.2 Job Seeker Profile Management

The job seeker profile schema contains personal and career related information related to Job Seekers. Structured information (education, work experience, skills, resumes) is maintained by related entities.

These records enable Job Seekers to have a complete professional profile which can be used to deliver relevant career information for job applications and resume functionality.

4.5.3 Employer and Company Management

The employer and company schema holds information needed for organizational and recruitment operations. Employers can manage information about their company and use the platform to carry out recruitment activities.

Company-related information is linked to recruitment workflows and job management operations, which helps employers manage company information and associated job opportunities.

4.5.4 Job Management

Job management schema provides information about job postings and related information. Job records are connected with the appropriate company and employer context.

The schema is designed to handle the lifecycle of a job, such as creation, publication, update, control and operations relating to the status of a given job. Job Seekers can view job information published based on the job browsing and application workflow of the application.

4.5.5 Job Application Management

The job application schema specifies the connections between Job Seekers and job postings. Each application represents a Job Seeker's application to a particular job.

Application information enables Employers to view applicants and track the recruitment status. The application process also allows the Job Seekers to track the applications they have submitted and the recruitment-related progress.

4.5.6 Resume and Career Information Management

The schema is used for resume-related information and structured content when creating a resume. Job Seeker profiles can be organized and utilized for related career-related information that can be used for resumes to be managed and application related workflows.

Information such as education, skills and professional experiences is relevant to the Job Seeker's professional profile and resume building.

4.5.7 Interview and Recruitment Workflow

The recruitment workflow schema can be used to facilitate activities between Employers and Job Seekers related to an interview. It helps to run the recruitment process after the application is received and assists in communicating information and answers about the interviews.

This architecture enables seamless integration of job applications to subsequent recruitment processes and offers structured handling of interview processes.

4.5.8 Notification Management

User notifications created by key system events are saved in the notification schema. Notifications are linked to their recipients, who can view relevant updates in the user interface via the application [1].

Database notifications are the primary persistent notification mechanism used in the platform. Stored database notifications are not dependent on optional real-time delivery mechanisms and can enhance the user experience when configured [1].

4.5.9 Support Ticket and Feedback Management

The support and feedback schema stores support requests and feedback submitted by users.

Support tickets help structure communication between users and administrators and handle platform-related issues or requests systematically. Feedback records capture comments, suggestions, or opinions related to the platform and its services.

4.5.10 Administrative and System Management

Administrative data structures enable operations at the Platform level, including user management, employer and company monitoring, job and application monitoring, reporting, Platform settings, Platform support operations, feedback management and other administrative operations.

These will enable authorized users to oversee and manage CareerConnectBD platform while having role-based access restrictions.

All the database schema is linked via well-defined relational relationships and Eloquent application relationships. The structure is used to maintain consistency of data and serves as the basis of the key functional modules of CareerConnectBD.

System Design

5.1 Component-Level Design Following Design Pattern(s)

CareerConnectBD is designed using a modular and layered approach based primarily on the Model-View-Controller (MVC) architectural pattern provided by the Laravel framework. The system separates data management, application logic, business operations, and user interface responsibilities into different components. This separation improves maintainability, testability, extensibility, and overall code organization.

In addition to the MVC pattern, the system uses service-layer components, middleware, form request validation, authorization policies, Eloquent ORM, and reusable frontend components where applicable. Figure 5.1 summarizes the major components and their typical interaction structure.

The major design components of CareerConnectBD are described below.

5.1.1 Presentation Layer

The presentation layer is responsible for displaying information to users and collecting user input. CareerConnectBD uses Vue.js for interactive and component-based frontend functionality. Bootstrap 5 is used for responsive styling, layouts, forms, navigation elements, cards, tables, buttons, modals, and other reusable user interface components.

Laravel Blade is also used for server-rendered views and layouts where applicable. Vite is used for managing and building JavaScript, Vue.js, and CSS assets.

The presentation layer provides separate interfaces according to the responsibilities of the three major user roles:

- Job Seeker Interface
- Employer Interface
- Admin Interface

Each role-specific interface provides access only to the features relevant to the authenticated user's role and permissions.

Component-Level Design of CareerConnectBD

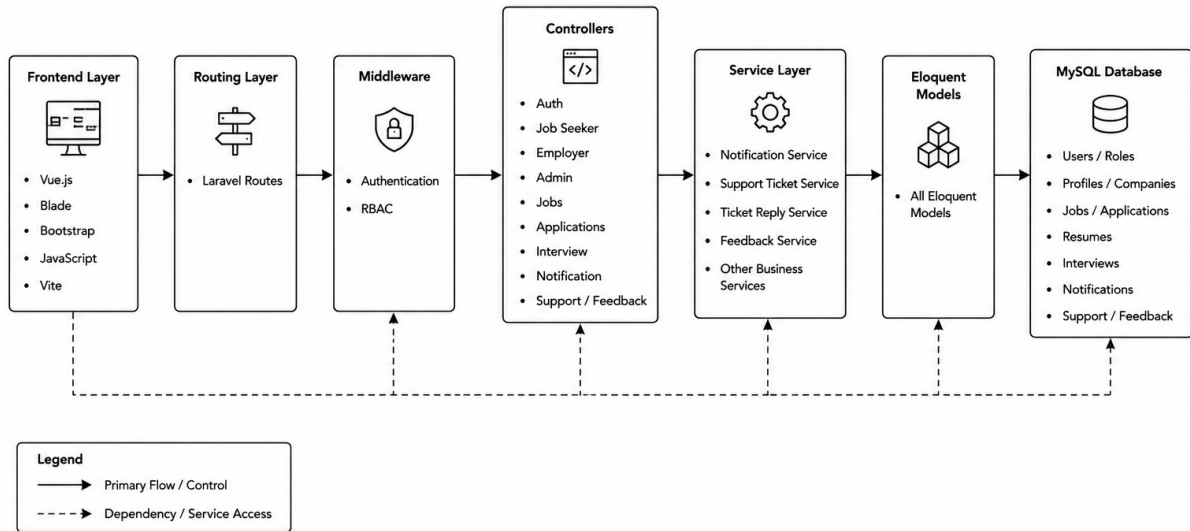


Figure 5.1: Component-Level Design of CareerConnectBD

5.1.2 Routing Layer

The routing layer defines the HTTP endpoints through which users interact with the application. Laravel routes connect incoming requests to the appropriate controllers or application actions.

Routes are organized according to public functionality and protected role-specific functionality. Authentication and authorization middleware are applied to protected routes to ensure that only authorized users can access specific system resources.

The routing structure supports major modules including authentication, Job Seeker operations, Employer operations, job management, application management, resume management, administrative operations, notifications, support tickets, and feedback.

5.1.3 Middleware and Authorization Layer

The middleware and authorization layer is responsible for controlling access to protected areas of the application.

Authentication middleware verifies whether a user is logged in before allowing access to protected resources. Role-based middleware verifies whether the authenticated user has the required role, such as Admin, Employer, or Job Seeker.

Authorization policies and ownership checks are used where applicable to prevent users from accessing or modifying resources that do not belong to them.

This design provides Role-Based Access Control (RBAC) and helps maintain security

boundaries among different categories of users.

5.1.4 Controller Layer

The controller layer receives HTTP requests from the routing layer and coordinates application operations.

Controllers are responsible for processing requests, invoking validation and authorization mechanisms, communicating with service classes and models, and returning appropriate responses or views.

CareerConnectBD uses controllers for major functional areas such as:

- Authentication and account management
- Job Seeker profile management
- Education, experience, and skill management
- Resume and resume builder management
- Job management
- Saved job management
- Job application management
- Employer and company management
- Interview and recruitment activities
- Administrative management
- Support and feedback operations

The controller layer is designed to coordinate requests while delegating selected complex business operations to dedicated service classes.

5.1.5 Service Layer

CareerConnectBD uses service-layer components for selected business operations that require logic beyond basic request processing.

Service classes help separate business rules from controllers and provide reusable operations for different parts of the application. Depending on the functional module, services are used to support administrative operations, notifications, support ticket processing, feedback handling, dashboard-related logic, and other business workflows.

This design reduces unnecessary complexity in controllers and improves the maintainability and testability of the application.

5.1.6 Validation Layer

The validation layer ensures that incoming user data satisfies the required application rules before it is processed or stored.

Laravel validation mechanisms and Form Request classes are used where applicable to validate data such as registration information, profile information, company information, job information, user-management information, and other application inputs.

Validation helps prevent incomplete, invalid, or improperly formatted data from entering the system.

5.1.7 Model and Data Access Layer

The Model layer represents the application's data entities and their relationships. Laravel Eloquent ORM is used to communicate with the MySQL relational database.

Models encapsulate database-related operations and define relationships among application entities. Major logical entities include users, roles, Job Seeker profiles, companies, jobs, job applications, education records, professional experiences, skills, resumes, resume builders, interview-related information, notifications, support tickets, feedback, and other supporting entities.

Eloquent relationships provide an organized method for retrieving related information and maintaining relationships between application records.

5.1.8 Authentication and Role-Based Access Control

CareerConnectBD uses Laravel's authentication mechanisms, with Laravel Breeze serving as the authentication foundation of the project.

The system supports role-based access for three major user categories:

- **Job Seeker:** Accesses career profile, education, experience, skills, resumes, resume builder, jobs, saved jobs, applications, interview-related functionality, notifications, support, and feedback features where available.
- **Employer:** Accesses company management, job posting and management, applicant review, candidate resumes, application status management, interview-related recruitment functionality, and notifications.
- **Admin:** Accesses the administrative dashboard, user management, platform monitoring, reporting, system settings, support ticket management, feedback management, and other administrative functionality.

Role-based middleware and authorization mechanisms prevent users from accessing functionality outside their assigned roles.

5.1.9 Notification Component

The notification component provides communication between different users based on important business events.

Database-backed notifications are used as the primary persistent notification mechanism. Relevant system activities can generate notifications for the appropriate recipients.

Examples of notification workflows include:

- An Employer receiving a notification when a Job Seeker submits an application to an applicable job.
- A Job Seeker receiving a notification when the status of an application changes.
- A Job Seeker receiving an interview-related notification.
- An Employer receiving information regarding relevant candidate interview responses.
- Administrators receiving relevant support ticket or feedback notifications.

Optional realtime broadcasting can be used as an enhancement when appropriately configured. The core notification design relies on persistent database notifications so that notification records remain available independently of realtime broadcasting availability.

5.1.10 File and Resume Management Component

The file and resume management component supports resume-related operations within CareerConnectBD.

Job Seekers can manage resume information and use the implemented resume builder functionality to prepare structured professional resumes. Resume-related operations may include creation, editing, previewing, printing, downloading, and managing uploaded resume files according to the available functionality.

Employers can access candidate resume information through authorized recruitment workflows where permitted by the application.

Access restrictions are applied to protect resume-related resources from unauthorized users.

5.1.11 Component Interaction

The major components of CareerConnectBD interact through a structured request-response workflow.

A typical request follows the sequence below:

1. A user interacts with the frontend through a web browser.
2. The presentation layer sends an HTTP request to the application.
3. Laravel routing identifies the appropriate route.
4. Middleware verifies authentication and role-based access.
5. The request is forwarded to the appropriate controller.
6. Validation and authorization are performed where required.
7. The controller invokes a service class or model operation.
8. Eloquent ORM communicates with the MySQL database.
9. The requested operation is processed.
10. The system returns the appropriate response to the presentation layer.

11. The updated information is displayed to the user.

5.2 GUI (Graphical User Interface) Design

The Graphical User Interface (GUI) of CareerConnectBD is designed to provide a structured, responsive, and role-oriented experience for Job Seekers, Employers, and Administrators.

The frontend interface uses Vue.js for interactive and component-based functionality, while Bootstrap 5 is used for responsive layouts and visual styling. Laravel Blade is also used for server-rendered views and layouts where applicable.

The GUI design follows a consistent visual structure across the application while providing different dashboards and navigation options according to the authenticated user's role.

5.2.1 Public Interface

The public interface provides access to general platform functionality that does not require authentication.

The public interface may include:

- Home or landing page
- Public job listings
- Job search and filtering
- Job details
- Login
- Job Seeker registration
- Employer registration

The public interface is designed to allow visitors to discover available opportunities and access the appropriate authentication or registration process.

5.2.2 Job Seeker Interface

The Job Seeker interface provides access to career management and job application functionality.

The Job Seeker dashboard presents relevant information and navigation options for features such as:

- Profile management
- Education management
- Professional experience management
- Skill management
- Resume management

- Resume builder
- Job browsing
- Saved jobs
- Job applications
- Application tracking
- Interview-related activities
- Notifications
- Support and feedback functionality

The interface is designed to help Job Seekers manage their professional information and recruitment activities from a centralized environment.

5.2.3 Employer Interface

The Employer interface is designed around company and recruitment management.

The Employer dashboard provides access to functionality such as:

- Company profile management
- Company-related information
- Job posting
- Job management
- Applicant management
- Candidate profile and resume review
- Application status management
- Interview-related recruitment activities
- Recruitment notifications

The Employer interface provides structured access to recruitment operations while restricting access to authorized company and job-related resources.

5.2.4 Admin Interface

The Admin interface provides centralized access to platform management and monitoring functionality.

The administrative interface includes access to implemented modules such as:

- Admin dashboard
- User management
- Employer and company monitoring
- Job monitoring

- Application monitoring
- Reports
- System settings
- Support ticket management
- Feedback management
- Administrative notifications

The Admin interface uses structured navigation, dashboard information, tables, forms, and management actions to provide administrators with centralized control over the platform.

5.2.5 Responsive Design

The CareerConnectBD interface is designed using responsive frontend techniques supported by Bootstrap 5.

Responsive layouts allow application pages and interface components to adapt to different common screen sizes. Components such as navigation menus, forms, cards, tables, buttons, and dashboard sections are organized to maintain usability across supported display sizes.

5.2.6 Consistency and Navigation

A consistent design approach is maintained throughout the platform to improve usability.

Role-specific navigation helps users identify available functionality. Common interface components such as navigation bars, sidebars, cards, forms, tables, buttons, dropdown menus, alerts, and notification indicators provide a familiar interaction pattern throughout the application.

The separation of Job Seeker, Employer, and Admin interfaces helps reduce unnecessary complexity by presenting users with functionality relevant to their assigned roles.

5.3 User Interface Screenshots

This section contains representative screenshots of CareerConnectBD user interfaces. The screenshots are intended to demonstrate selected public pages, authentication screens, and role-specific workflows.

5.3.1 Public Interface Screenshots

The following figures provide multiple views of the public-facing pages to document the full layout and content presented to unauthenticated users.

Home Page

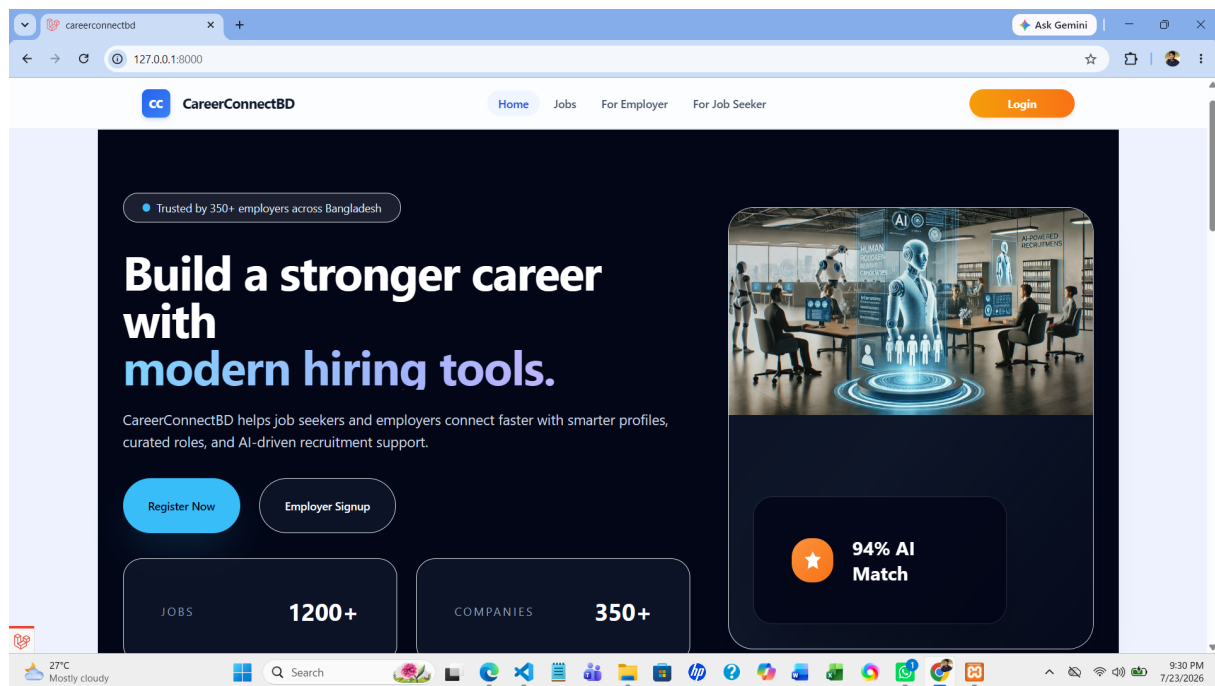


Figure 5.2: CareerConnectBD Home Page.

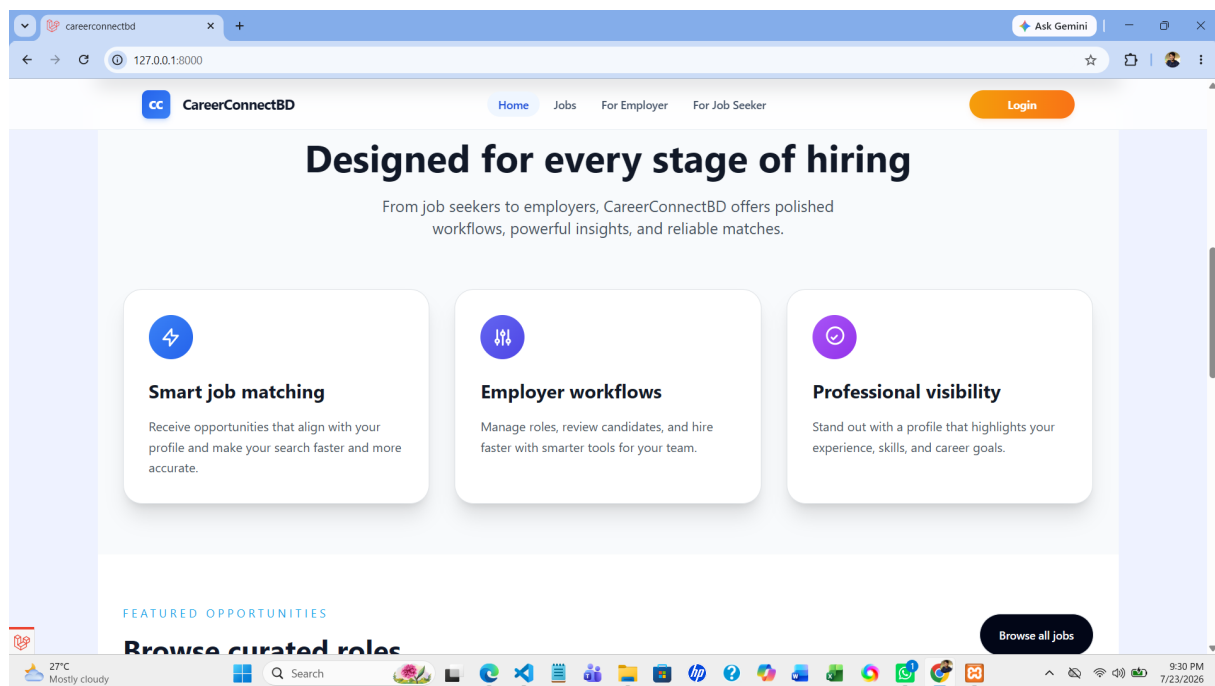


Figure 5.3: CareerConnectBD Home Page – Additional Content I.

Job Listing

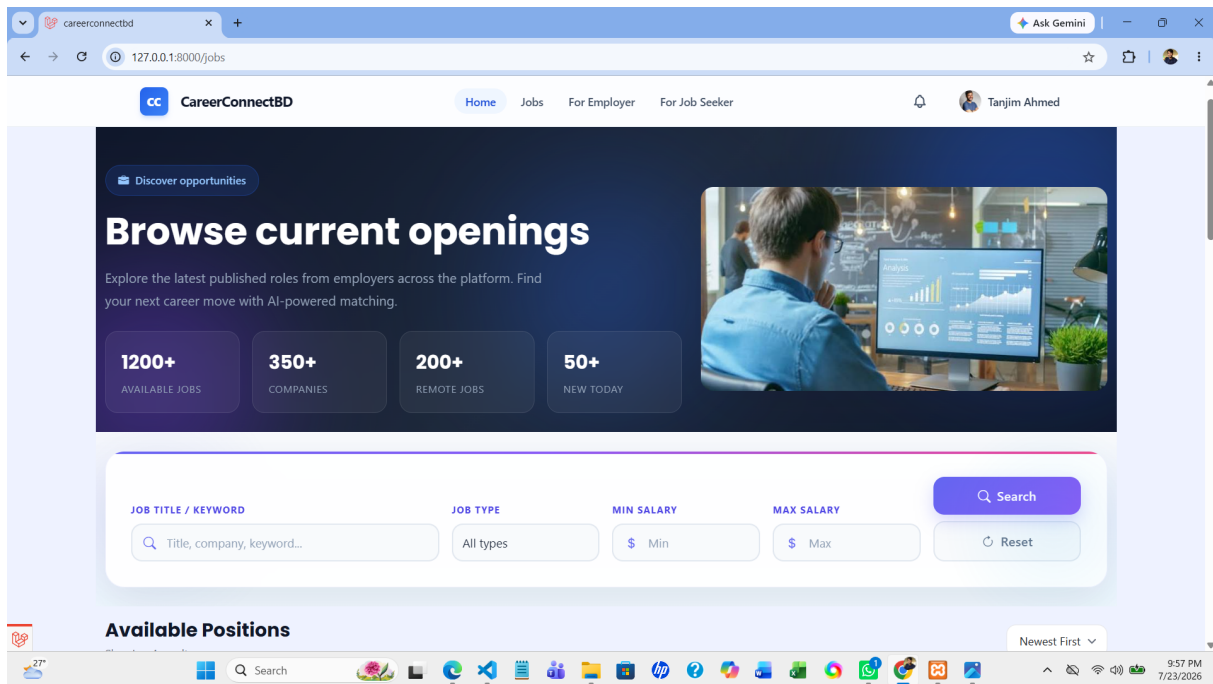


Figure 5.4: Job Listing Interface – Primary View.

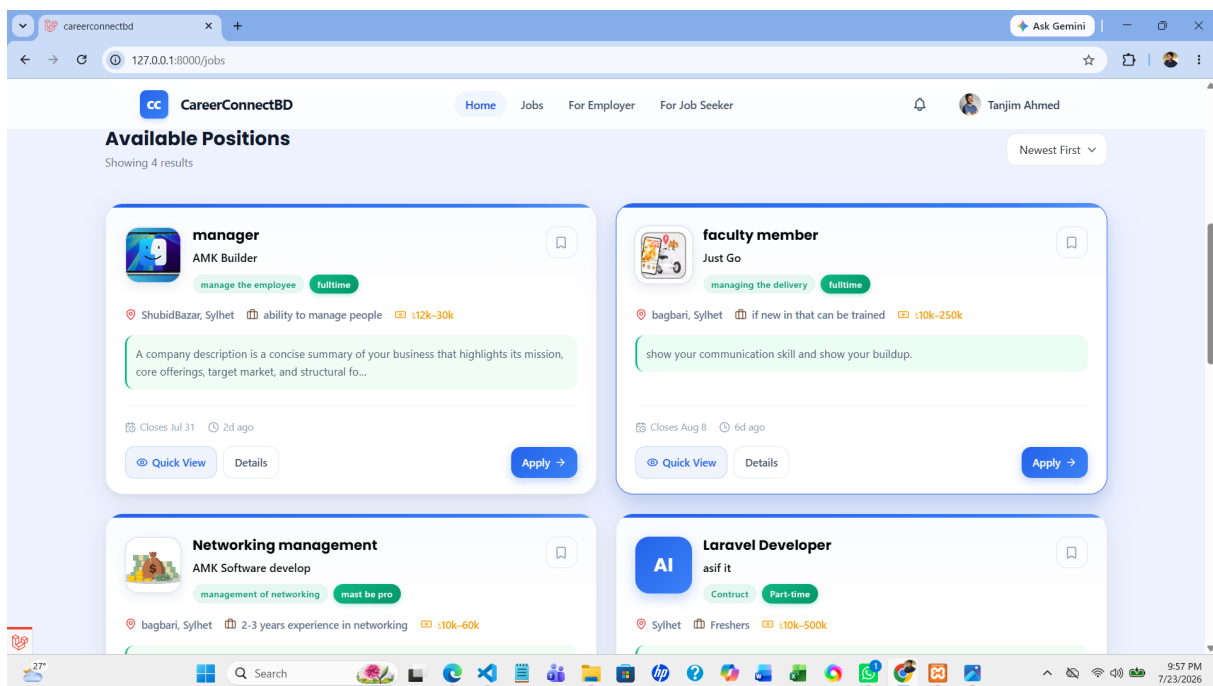


Figure 5.5: Job Listing Interface – Additional View.

Job Details

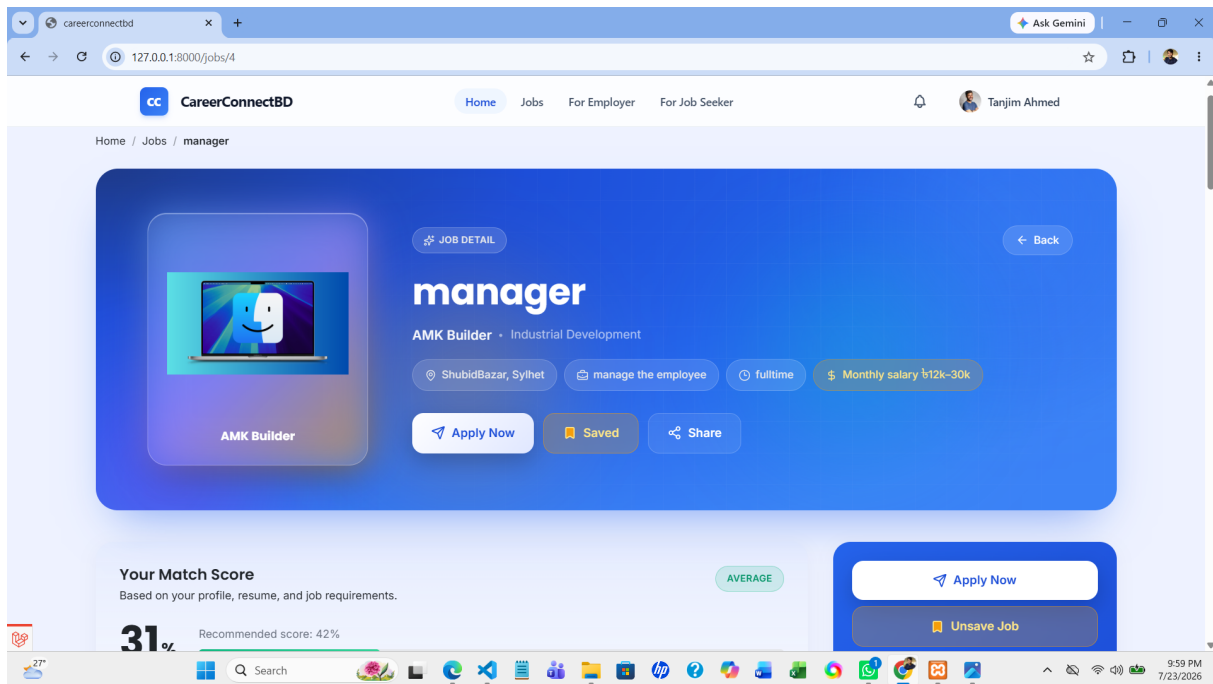


Figure 5.6: Job Details Interface – Overview.

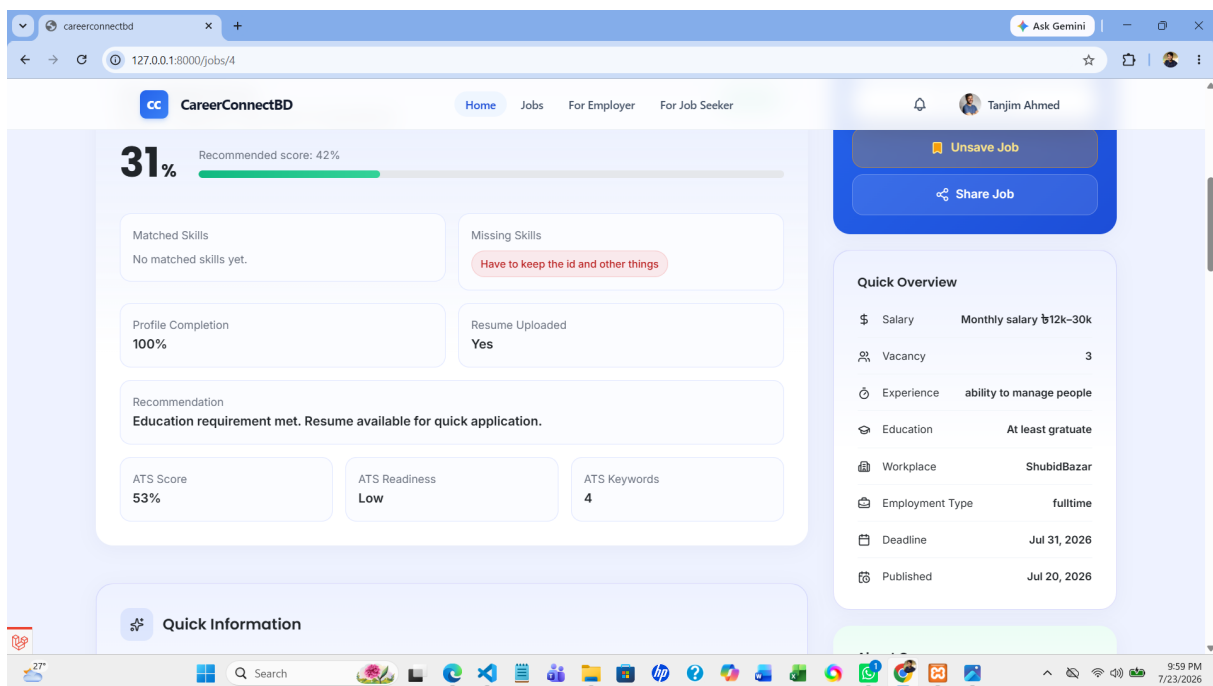


Figure 5.7: Job Details Interface – Additional Information I.

5.3.2 Authentication Interface Screenshots

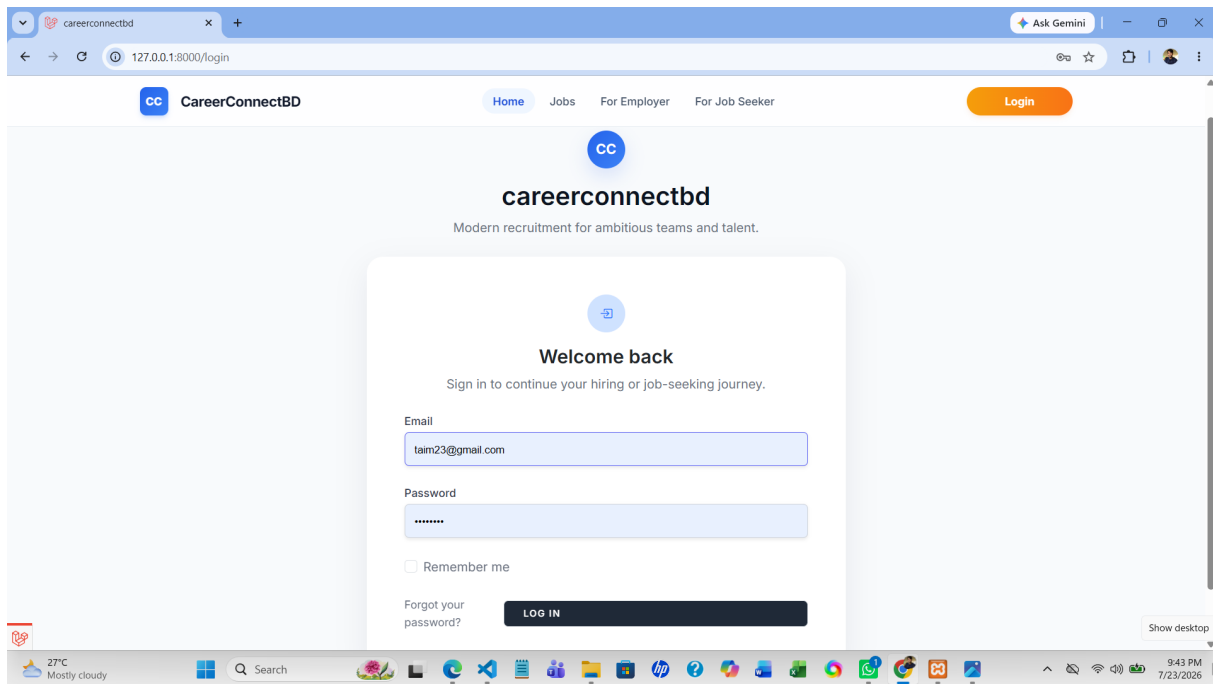


Figure 5.8: Login page for authenticated access.

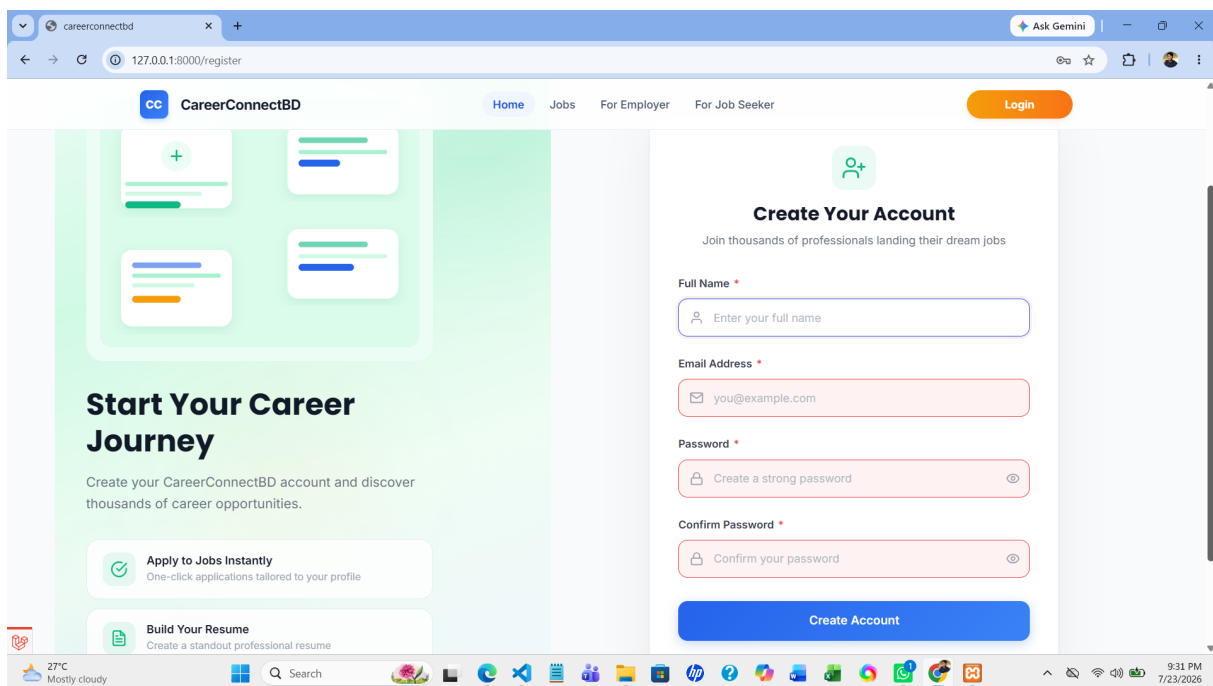


Figure 5.9: Job seeker registration page for creating a candidate account.

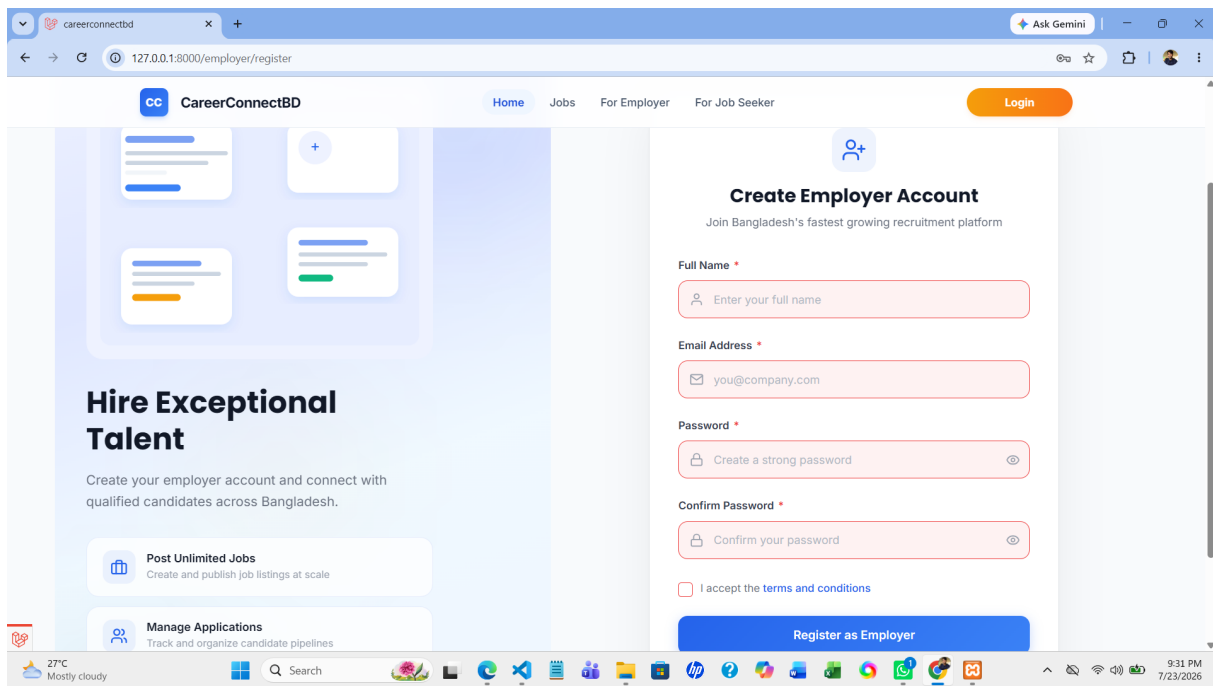


Figure 5.10: Employer registration page for onboarding recruiting organizations.

5.3.3 Job Seeker Interface Screenshots

Job Seeker Dashboard

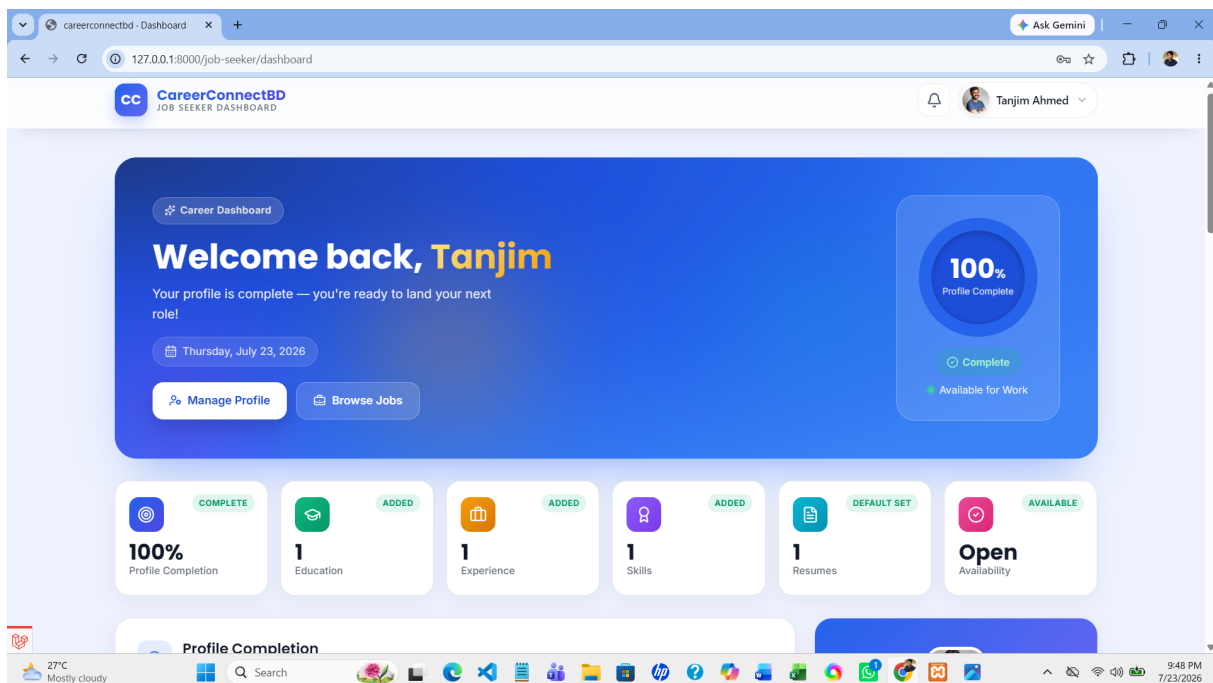


Figure 5.11: Job seeker dashboard summarizing activity and available actions.

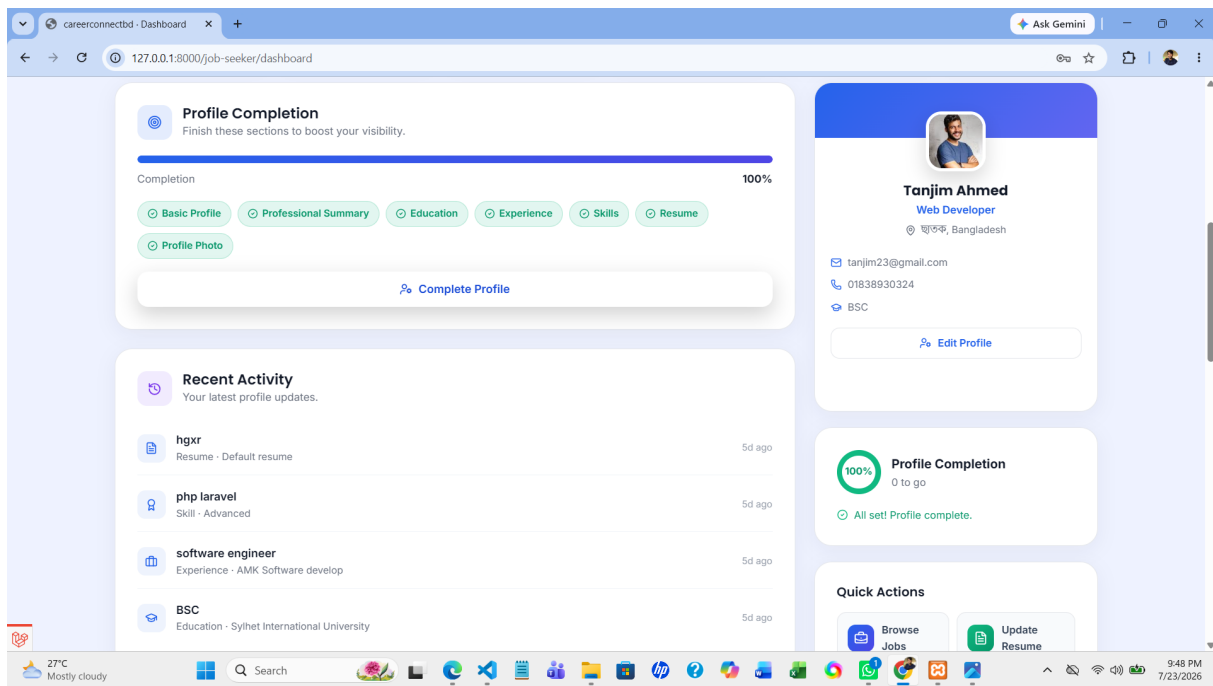


Figure 5.12: Job seeker dashboard – additional view I.

Resume Management

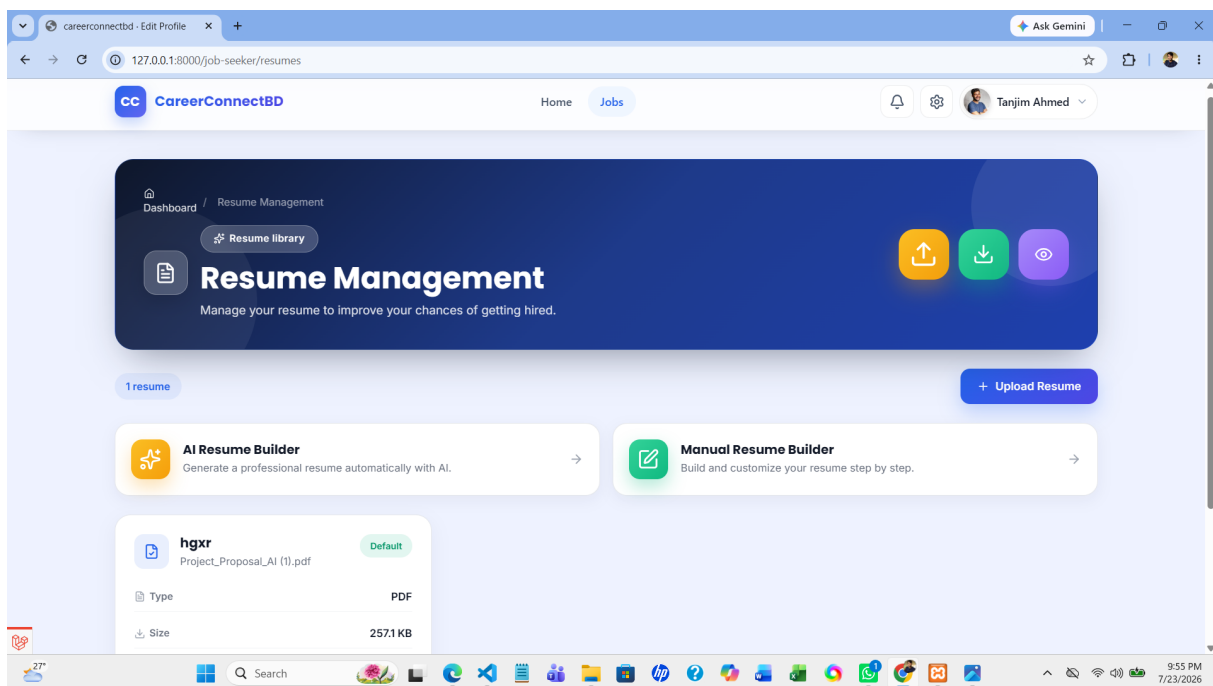


Figure 5.13: Resume management interface for uploading and organizing candidate documents.

Job Application

The screenshot shows a web browser window with the URL `127.0.0.1:8000/job-applications/create?job_id=4`. The page header includes the CareerConnectBD logo and navigation links for Home, Jobs, For Employer, and For Job Seeker. The user is logged in as Tanjim Ahmed. The main content area is titled "Apply for This Job" and includes a sub-header "Your Application". Below this, there are two sections: "Application Details" and "Job Summary". The "Application Details" section contains a "Resume" field. The "Job Summary" section displays the company name "CareerConnectBD" and the position. A large, semi-transparent form overlay is visible in the background, suggesting a draft or a preview of the application form.

Figure 5.14: Job application form used to submit candidate applications.

This screenshot shows the same web browser window, but with the "Application Details" section expanded. It features a "Select your resume" dropdown menu, an "UPLOAD FROM YOUR DEVICE" section with a "Click to choose a file" button, and a "Cover Letter" text area. The "Cover Letter" section includes a placeholder text "Tell the employer why you are a great fit..." and a character count "Minimum 50 characters; maximum 3000 characters." At the bottom, there are "Back to Job" and "Submit Application" buttons. The "Job Summary" section on the right remains visible, showing the company and position details. An "Application Checklist" is also present, listing items like "Resume attached", "Profile completed", "Review your information", and "Double-check before submission".

Figure 5.15: Job application interface – additional view.

Application Tracking

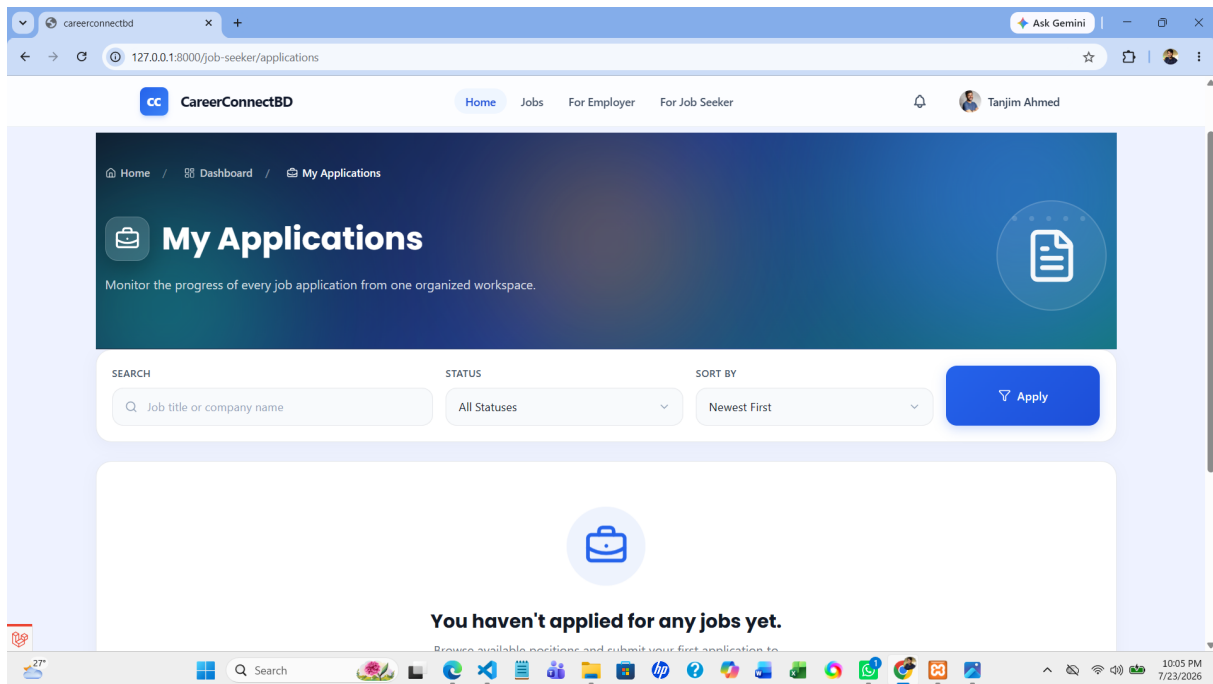


Figure 5.16: My applications page showing submitted applications and their statuses.

5.3.4 Employer Interface Screenshots

The following screenshots present the major Employer-facing interfaces of CareerConnectBD. These interfaces support employer dashboard access, company profile management, job management, and applicant management within the recruitment workflow.

Employer Dashboard

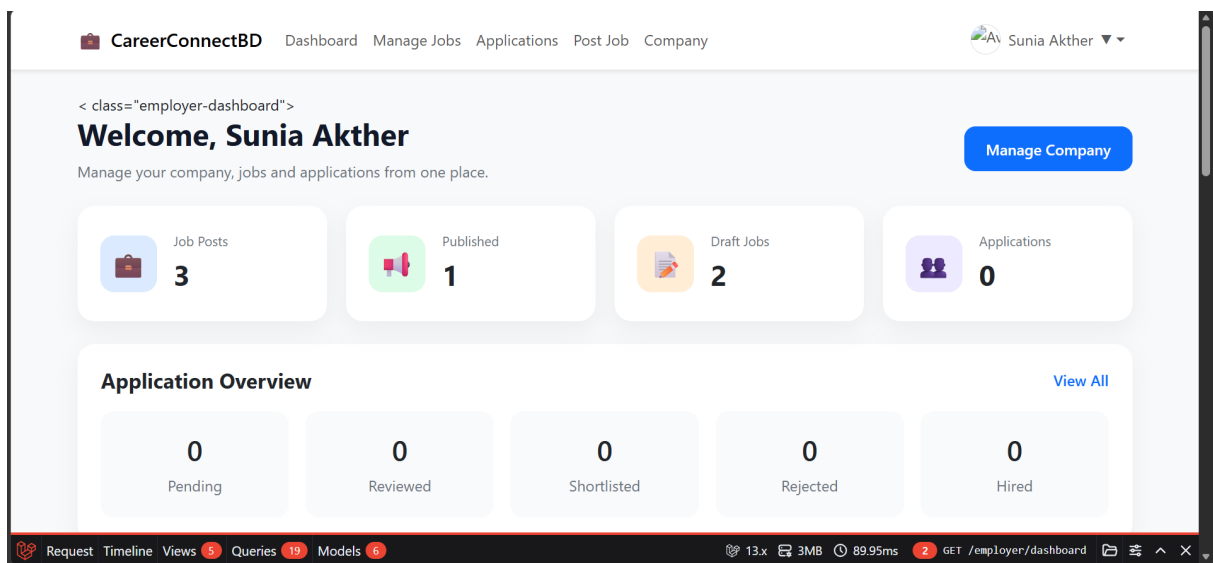


Figure 5.17: Employer Dashboard of CareerConnectBD.

Company Profile Management

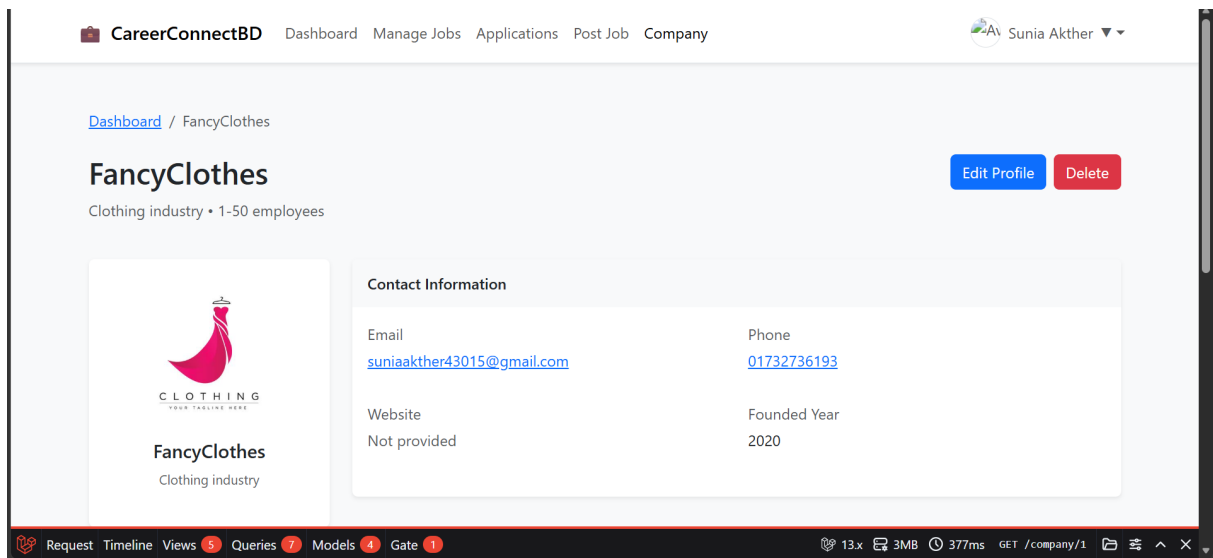


Figure 5.18: Employer Company Profile Management Interface.

Employer Job Management

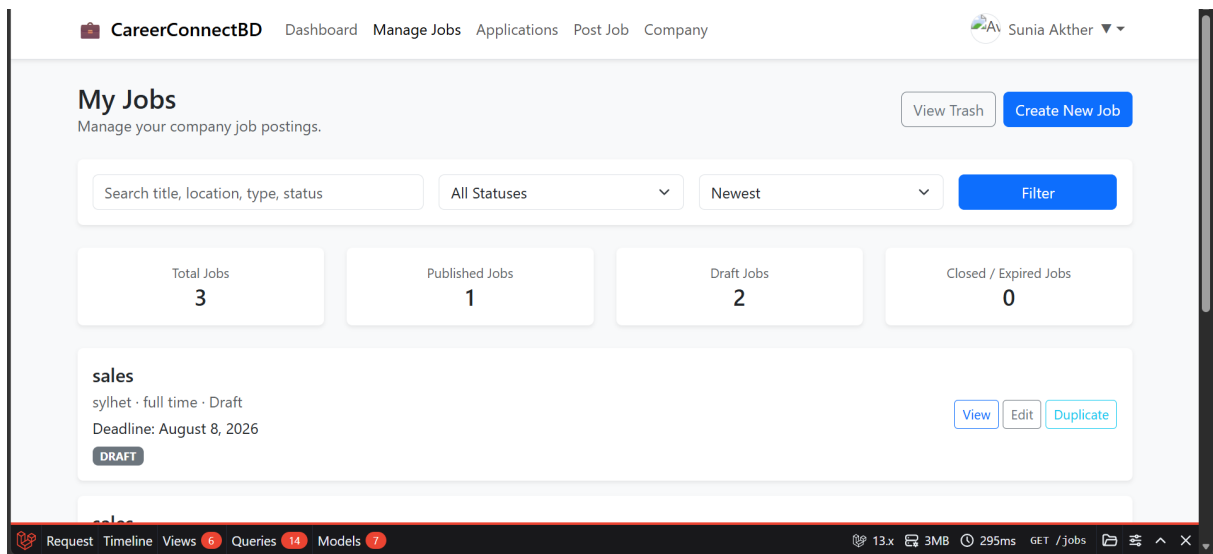


Figure 5.19: Employer Job Management Interface.

Applicant Management

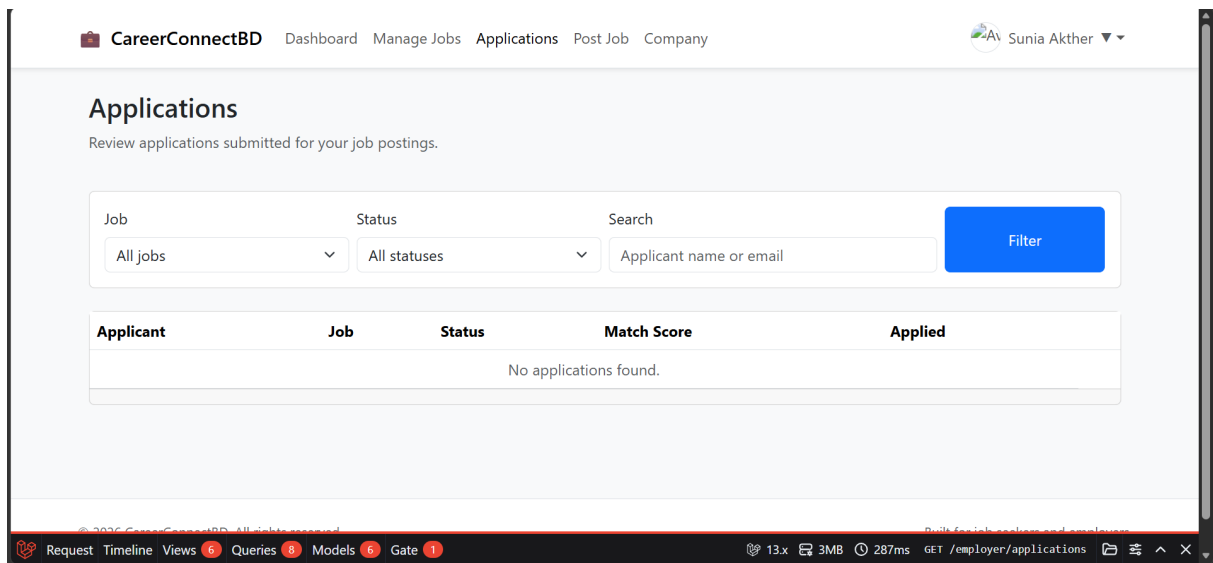


Figure 5.20: Employer Applicant Management Interface.

5.3.5 Admin Interface Screenshots

Admin Dashboard

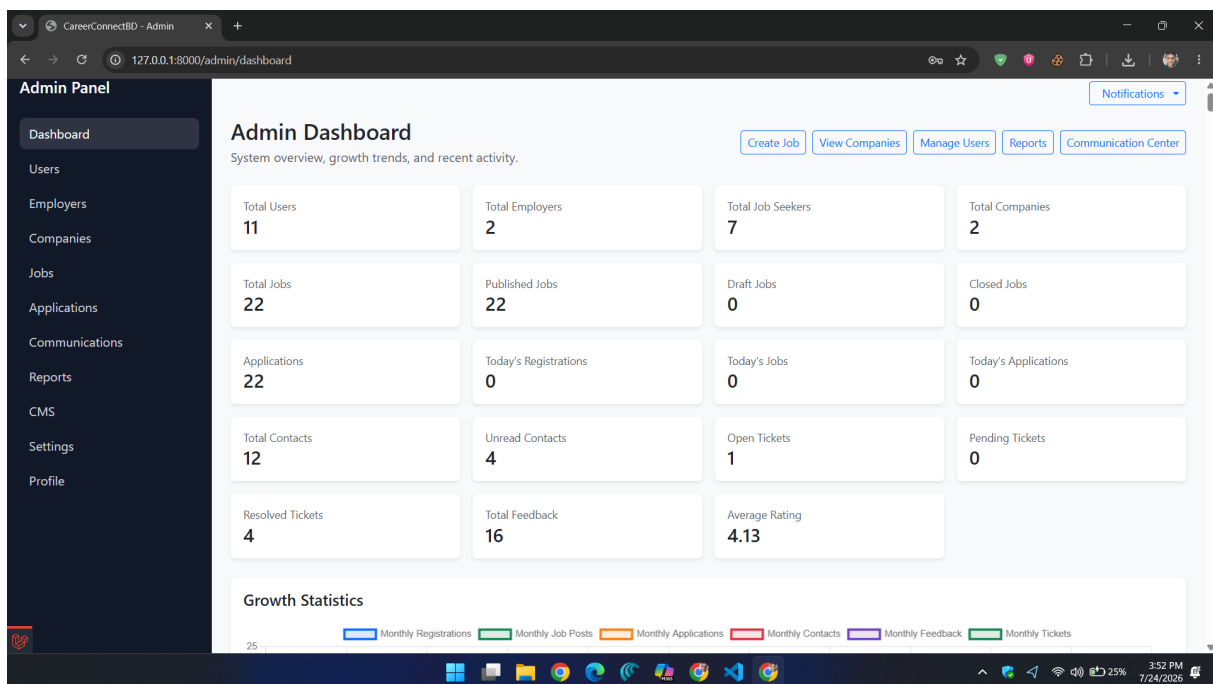


Figure 5.21: Admin Dashboard Interface.

System Implementation

6.1 Development Environment Setup

The CareerConnectBD development environment was set up to provide backend and frontend development support, database management, testing, and version control functionality. The application has been built with Laravel 13 as the main backend framework [1] and PHP 8.4 as the server-side programming language [3]. A relational database management system, called MySQL, was used.

On the frontend, Vue.js is employed to create interactive user interface features as well as the functionality of the components, and Bootstrap 5 is utilized for responsive design and user interface components. Where applicable, server rendered views and layouts are also created using Laravel Blade. Vite is a tool for managing and building JavaScript, Vue.js and CSS assets.

The main app development environment was Visual Studio Code. The source code was versioned using git, and the repository was hosted on GitHub and developed collaboratively by the members of the project team.

The key technologies and tools employed in the development environment are:

- Laravel 13
- PHP 8.4
- Vue.js
- JavaScript
- Bootstrap 5
- Laravel Blade
- Vite
- MySQL
- Composer
- Node.js and NPM
- Visual Studio Code
- Git

- GitHub

Frontend dependencies are managed with NPM and the Laravel project dependencies are managed with Composer [9]. The required database structure and initial system data is created and initialized using Laravel migrations and seeders.

The application can be started from the processes of installing the required dependencies, configuring the application's environment, database migration, and asset build process for the frontend application. Database operations, cache management, application testing and local server execution are all done through Laravel Artisan commands.

The development is done in a structured way, where backend functions are added, and then interfaces, database schemas, authentication and authorization, business logic, and finally testing.

6.2 Module-wise Implementation

CareerConnectBD is implemented as a modular application, whereby the major functional requirements are broken down into logical modules. Each module deals with one particular aspect of the career and recruitment management process.

6.2.1 Authentication and Role Management Module

The authentication module offers password-related functionality, user registration, login, logout and authenticated session management. Laravel breeze is the authentication system of the system.

CareerConnectBD supports three primary roles:

- Admin
- Employer
- Job Seeker

Role-based access control is implemented to ensure that users can access only the functionality associated with their assigned roles. Middleware and authorization mechanisms protect role-specific routes and system resources.

After authentication, users are directed to the appropriate system area according to their role and account context.

6.2.2 Job Seeker Profile Module

The Job Seeker Profile module allows Job Seekers to maintain their personal and professional information.

The module supports structured career information including:

- Personal profile information
- Educational qualifications
- Professional experience

- Skills
- Resume-related information

Job Seekers are able to build and keep their profile data using secure interfaces. The authenticated Job Seeker is linked to related records to ensure proper ownership and data isolation.

Education, Experience and Skill Management Module

Educational Qualifications, Professional Experience and Skills are managed individually in CareerConnectBD.

Based on the implemented functionality, Job Seekers are able to make, view, change and delete their career related records. These records help to build a user's professional profile and can help in resume creation and recruitment work.

6.2.3 Resume Management Module

The Resume Management module enables Job Seekers to keep resume information and files based on the workflow that has been adopted.

The system offers resumes management functionality and availability of proper resumes information at the job application and recruitment stages.

Authorized Employers have access to candidate resume information via appropriate application workflows as allowed by the system. To stop unauthorized access to resume related resources they use access control mechanism.

6.2.4 Resume Builder Module

The Resume Builder module offers Job Seekers features to help build a professional, structured resume in CareerConnectBD.

The workflow implemented can assist with operations like:

- Creating a resume
- Editing resume information
- Previewing a generated resume
- Printing a resume
- Downloading supported resume output

The Resume Builder integrates career-related information and structured resume sections to help Job Seekers maintain professional documents within the platform.

6.2.5 Job Management Module

The Job Management module allows Employers to create and manage job opportunities associated with their recruitment activities.

Depending on the implemented job lifecycle, Employers can perform operations such as:

- Creating job postings
- Editing existing jobs
- Managing job information
- Deleting or soft-deleting jobs
- Restoring eligible deleted jobs
- Duplicating job postings

Publicly available job information can be accessed through the job listing and job details interfaces. Job Seekers can browse available opportunities and proceed with relevant job-seeking actions.

6.2.6 Saved Job Module

The Saved Job module allows authenticated Job Seekers to save preferred jobs for later access.

A Job Seeker can add an available job to the saved job collection and access saved opportunities through the appropriate interface. This functionality helps users organize jobs of interest before deciding whether to submit an application.

6.2.7 Job Application Module

The Job Application module manages the interaction between Job Seekers and available job opportunities.

Authenticated Job Seekers can submit applications for eligible jobs. Application records connect the applicant with the corresponding job and provide the foundation for subsequent recruitment activities.

The module supports application-related functionality including:

- Job application submission
- Application record management
- Duplicate application prevention where applicable
- Job Seeker application tracking
- Employer applicant review
- Application status management

Employers can access applications associated with their recruitment context and review relevant candidate information.

6.2.8 Employer and Company Management Module

The Employer module offers tools for managing the activities of recruitment and information of the company.

Through a secure dashboard, employers can keep on the company information and gain access to the employer specific functionality.

The module combines company management and job posting management with applicant management, to create a unified space for the organization's recruitment activities.

6.2.9 Applicant Management Module

The Applicant Management module enables the Employer to view the Job Seekers who have applied for job(s) suitable to them.

Employer access to application information and candidate data on authorization basis. Where supported by application, resume preview/ Download is available.

Candidates' applications status can be also be updated by the employer as they proceed through the recruitment process.

6.2.10 Interview and Recruitment Module

The Interview and Recruitment module is an extension of the application process that facilitates interview activities between Employers and Job Seekers.

The recruitment workflow implemented helps with the relevant operations like sending out invite to Interview or response from the candidate, wherever applicable.

Actions that happen during the interview are connected to notification system so that appropriate users can be notified about important recruitment actions.

6.2.11 Notification Module

Notice is implemented using a database in CareerConnectBD.

The notification module will deliver persistent notifications for activities that are relevant to the business. Examples include:

- Notification to an Employer when a Job Seeker submits a relevant job application.
- Notification to a Job Seeker when an application status changes.
- Notification to a Job Seeker for relevant interview invitations.
- Notification to an Employer for relevant candidate interview responses.
- Notification to Administrators for relevant support tickets.
- Notification to Administrators for relevant feedback submissions.
- Notification to ticket owners for applicable administrative replies.

Database notifications serve as the primary source of persistent notification information. Optional realtime broadcasting can be enabled as an enhancement when the required realtime infrastructure is available.

The notification architecture is designed so that failure or unavailability of optional realtime broadcasting does not prevent the core database notification process from operating.

6.2.12 Admin Management Module

The Admin Management module provides authorized Administrators with centralized access to platform management functionality.

The administrative environment includes functionality for major operations such as:

- Administrative dashboard
- User management
- User status management
- Employer monitoring
- Company monitoring
- Job monitoring
- Application monitoring
- Reports
- System settings
- Support ticket management
- Feedback management

Administrative routes are secured via authentication and role based access control to ensure that they are not accessible by unprivileged users in accessing administrative functionality.

6.2.13 Support Ticket Module

The Support Ticket module allow users to send an organized ticket to the Administrators regarding any platform related issue or request.

It is possible to submit support tickets using the support workflow provided. Administrators can check the appropriate tickets and respond to them.

Where appropriate, support ticket activities are tied to the notification architecture so that appropriate people are notified about ticket related activities.

6.2.14 Feedback Module

The Feedback module is for users to provide feedback about the CareerConnectBD platform.

Feedback submitted can be viewed by administrators at the Administrative interface. Feedback related notifications can be used to alert appropriate Administrators of new feedback submitted based on the workflow implemented.

6.2.15 Reporting and System Settings Module

Administrative reporting offers platform-level information and summary reports to authorized Administrators.

The System Settings module offers administrative access to platform information that can be configured (supported by the implementation).

These modules enable Administrators to keep an eye on activities on the platform and ensure the supported application configuration.

6.3 Key Features Implementation

CareerConnectBD has a number of significant features that assist in making it one environment for career and recruitment management.

6.3.1 Role-Based Dashboard

The system has individual dashboards and role specific navigation for Job Seekers, Employers and Administrators.

The authentication is then used to check user's role and access the corresponding functionality of the application. This design minimises superfluous interface complexity and no unauthorised access to role specific operations.

6.3.2 Integrated Career Profile Management

Job Seekers are able to have several types of professional information in one profile.

Education, experience, skills and resume details are organized information that helps to provide a more organized career profile management.

6.3.3 Resume Builder

A built-in Resume Builder in the feature for Job Seekers allows them to create and maintain their resumes without the need for an outside resume builder.

The feature will allow for resumes to be created and managed, such as editing, previewing, printing, and supported download.

6.3.4 Job Discovery and Saved Jobs

Job discovery function will give access to job opportunity available along with in-depth information about the job.

Job Seekers can find jobs of interest, and store them in their own job bank for later viewing. This will make it easier for a person to search for a job.

6.3.5 Application Tracking

CareerConnectBD offers an integrated application workflow, which links Job Seekers and Employers together.

Job Seekers can apply and monitor the data relevant to their applications and Employers can view applicants and update the status of the recruitment.

This offers a structured recruitment procedure in the platform.

6.3.6 Candidate Review and Recruitment Workflow

Employers can use protected recruitment interfaces to view applications from candidates and only choose those they consider suitable, and to view information about candidates.

Application status management and Interview related functionality helps to take the recruitment process forward beyond Job applications.

6.3.7 Notification Workflow

The notification system links crucial business actions to the right people.

Important notification are stored in the database for persistence, and not just relying on transient frontend message. This allows the users to obtain information about the notification with the application.

The delivery is the primary method but if set up, optional realtime broadcasting can provide a more effective notification delivery mechanism.

6.3.8 Administrative Control

Oversight of the platform from a centralized platform in the administrative environment.

Authorised Administrators control Users and monitor the important assets on the platform. There is a feature to write feedback and a support ticket, system settings, and reporting on the platform administration.

6.3.9 Support and Feedback

CareerConnectBD has designated user feedback and support requests.

The support ticket system promotes a controlled interaction between Administrators and users, while the feedback system provides an opportunity for users to comment and make suggestions for the platform.

6.4 Security and Validation Mechanisms

Since CareerConnectBD handles professional information, user accounts, resumes, job applications, company information and recruitment related data, security is an important aspect.

Various layers of security and validation provided by Laravel are added to the application, as well as the deployment of the application architecture.

6.4.1 Authentication

For registered users, authentication mechanisms are used to check before using protected functions of the system.

In Laravel Breeze, the registration, authentication and logins are provided by the package, so are password functionality and authenticated sessions.

6.4.2 Role-Based Authorization

Admin, Employer and Job Seeker Users are separated by using Role Based Access Control.

To ensure that only users with the proper role can be passed to the appropriate route groups, role middleware ensures that the user is assigned the appropriate role. Resources that are specific to each role can only be accessed by the users that have been authorized.

authorization Policies and Ownership Protection

Restrictions on authorisation and ownership are in place as appropriate to ensure the protection of sensitive resources.

For example, if the user is authorized to access records for his or her own account and/or for recruiting within that account, then only the records belonging to that account should be visible to the user. This will help to prevent unauthorized access to other user's resources or making changes to other users' resources without their consent.

6.4.3 Form Request Validation

Checks user input to ensure that it is acceptable and can be processed or stored.

Laravel validation mechanisms or specific Form Request classes are used as necessary to validate the data according to the application needs.

Validation helps to minimize the possibility of entering incomplete, unformed or inaccurate data into the system.

6.4.4 Password Security

Laravel's built-in, secure password hashing functions are used to store user passwords, not plain text.

When the user types in credentials during the authentication process, their credentials will be checked against the authentication framework.

6.4.5 CSRF Protection

If applicable, state changing web requests are protected against Cross-Site Request Forgery (CSRF) using Laravel's CSRF protection.

CSRF tokens can be used to make sure that requests sent via forms requiring protected sessions are from legitimate application sessions.

6.4.6 Database Integrity

Primary keys, foreign keys and consistency of relationships between related records are defined by the relational database structure.

Data integrity is added at the application level and Eloquent relationship level to maintain it across the platform.

6.4.7 Protected File Access and Resume Access

Authorized application workflows are used to access resume-related resources and resume information.

Employers have limitations regarding who they are allowed to recruit and Job Seekers control their resume and information about their careers based on the rules of what they are allowed to apply for.

6.4.8 Administrative Protection

Authentication and a special “Admin” role authorization are used for administrative route security.

When appropriate, extra administrative safeguards are used to minimize the risk of inappropriate administrative action in sensitive user-management operations.

6.4.9 Notification Reliability

Database based notifications is the major notification mechanism.

Routine live streaming is regarded as an add-on feature, and not a necessity for the basic operation of business. This design can be used to guarantee that the records of database notifications or key application workflows aren’t blocked when there is a temporary issue with the realtime service.

6.5 Deployment

The CareerConnectBD is currently running and testing it in a **local development environment** (localhost) during the project implementation and verification phase. This project state does not have any public production deployments.

At the same time, the system has been designed based on the standard design of the application Laravel that can be deployed in a suitable system in the future. To run the application in production, you must have a web server (such as Apache or Nginx) and PHP versions that are supported, Composer versions that are supported, a MySQL database, and Node and NPM versions that are supported to build frontend assets with Vite.

The normal deployment process should consist of the following steps: Copying the source code, installing the **Composer** and **NPM** dependencies, setting up the `.env` file, migrating the database, building the application for production on the front end (`npm run build`), configuring the storage permissions, and configuring the web-server (`public` directory of Laravel).

The notification workflow works without the need for optional realtime broadcasting, and using persistent database notifications. The realtime delivery capability is to be activated in future deployments but can be deployed as an add-on service rather than a fundamental business service.

Testing and Sustainability Plan

7.1 Requirements/specifications-based System Level Test Cases

System-level testing was performed to verify whether the major functional requirements of CareerConnectBD operate correctly from the perspective of different system users. The testing process focused on authentication, role-based authorization, job seeker activities, employer operations, job applications, resume management, recruitment workflows, notifications, administrative functions, and support and feedback management.

The major requirements considered for system-level testing are summarized in Table 7.1.

Table 7.1: Requirements/specifications-based system-level requirements

Requirement ID	Requirement Statement	Must/Want	Comment
R-AUTH-01	Registered users can log in using valid credentials and are redirected according to their assigned roles.	Must	Core authentication
R-AUTH-02	Unauthorized users cannot access protected resources belonging to other roles.	Must	RBAC security
R-JS-01	A Job Seeker can manage profile information and career-related information.	Must	Job Seeker module
R-JOB-01	A Job Seeker can browse available jobs and submit a valid job application.	Must	Job application
R-EMP-01	An Employer can manage company information and authorized recruitment resources.	Must	Employer module
R-JOB-02	An authorized Employer can create and manage job postings.	Must	Job management
R-APP-01	An Employer can view applications associated with authorized job postings and update application status.	Must	Application management
R-RES-01	A Job Seeker can manage resume-related information using the available resume management and resume builder features.	Must	Resume management
R-INT-01	The recruitment workflow supports interview invitation and candidate response operations.	Must	Interview management

Requirement ID	Requirement Statement	Must/Want	Comment
R-NOT-01	Important system activities generate notifications for the intended recipient.	Must	Notification system
R-ADM-01	An Admin can access protected administrative modules and perform authorized platform management operations.	Must	Administration
R-SUP-01	Users can submit support requests and the administrative workflow can manage the corresponding support communication.	Must	Support system
R-FBK-01	Users can submit feedback through the platform for administrative review.	Want	Feedback management

7.1.1 Authentication and Authorization Test Cases

Table 7.2: Authentication and authorization system-level test cases

Test Case ID	Scenario	Steps	Test Data	Expected Result	Actual Result	Status
TC-AUTH-01	Login with valid credentials	Open login page; enter valid credentials; submit login form	Valid registered account	User is authenticated and redirected to the appropriate role-based area	User is authenticated and redirected according to role	Passed
TC-AUTH-02	Login with invalid credentials	Enter registered email with incorrect password and submit	Invalid password	Authentication is rejected and an appropriate validation response is shown	Authentication is rejected	Passed
TC-AUTH-03	Access protected route as guest	Open a protected dashboard URL without authentication	Guest session	User is redirected to the login page	Guest is redirected to login	Passed
TC-AUTH-04	Access unauthorized role resource	Login with one role and attempt to access a protected resource of another role	Authenticated unauthorized user	Access is denied by authorization mechanism	Unauthorized access is denied	Passed

7.1.2 Job and Application Management Test Cases

Table 7.3: Job and application management system-level test cases

Test Case ID	Scenario	Steps	Test Data	Expected Result	Actual Result	Status
TC-JOB-01	Browse available jobs	Open the job listing page and view available job information	Available job records	System displays available job information	Available jobs are displayed	Passed
TC-JOB-02	Employer manages a job posting	Login as Employer; open job management; perform an authorized job operation	Valid job information	Job operation is processed successfully	Job operation completes successfully	Passed
TC-APP-01	Job Seeker submits application	Login as Job Seeker; select a job; submit a valid application	Valid job and applicant	Application is stored and associated with the correct job and applicant	Application is created successfully	Passed
TC-APP-02	Prevent duplicate application	Attempt to apply again to a job for which an application already exists	Existing application	Duplicate application is rejected	Duplicate submission is prevented	Passed
TC-APP-03	Employer updates application status	Login as authorized Employer; open application; change status	Valid application and status	Application status is updated successfully	Application status is updated	Passed

7.1.3 Resume, Interview, and Recruitment Test Cases

Table 7.4: Resume and recruitment workflow system-level test cases

Test Case ID	Scenario	Steps	Test Data	Expected Result	Actual Result	Status
TC-RES-01	Manage resume information	Login as Job Seeker; access resume feature; provide valid resume-related data	Valid resume data	Resume information is processed and associated with the Job Seeker	Resume information is managed successfully	Passed

Test Case ID	Scenario	Steps	Test Data	Expected Result	Actual Result	Status
TC-INT-01	Send interview invitation	Authorized Employer selects an eligible application and creates an interview invitation	Valid application and interview information	Interview invitation is created for the intended candidate	Invitation is created successfully	Passed
TC-INT-02	Candidate responds to invitation	Job Seeker opens interview invitation and submits a valid response	Accept or decline response	Response is stored for the correct interview invitation	Interview response is stored successfully	Passed

7.1.4 Notification, Administration, and Support Test Cases

Table 7.5: Notification, administration, and support system-level test cases

Test Case ID	Scenario	Steps	Test Data	Expected Result	Actual Result	Status
TC-NOT-01	Generate notification for job application	Job Seeker successfully submits an application	Valid application	Notification is created for the responsible Employer	Employer notification is created	Passed
TC-NOT-02	Notify Job Seeker of status change	Employer changes an application's status	Updated application status	Notification is created for the corresponding Job Seeker	Job Seeker notification is created	Passed
TC-NOT-03	Notification works without realtime server	Perform a notification-generating business action while realtime broadcasting is unavailable	Realtime disabled or unavailable	Database notification is retained and core business request succeeds	Core request and database notification remain functional	Passed
TC-ADM-01	Admin accesses administrative module	Login as Admin and open protected admin area	Valid Admin account	Admin dashboard and authorized management modules are accessible	Authorized admin resources are accessible	Passed

Test Case ID	Scenario	Steps	Test Data	Expected Result	Actual Result	Status
TC-SUP-01	Submit support ticket	Authenticated user opens support interface and submits valid ticket information	Valid ticket data	Support ticket is stored and made available to the administrative workflow	Support ticket is created successfully	Passed
TC-FBK-01	Submit platform feedback	User submits valid feedback through the available feedback feature	Valid feedback	Feedback is stored for administrative review	Feedback is recorded successfully	Passed

7.2 Traceability of Test Cases to Use Cases

The traceability matrix maps the major system requirements to representative system-level test cases. This ensures that important functional requirements are covered by at least one corresponding test scenario.

Requirement ID	Related Test Case(s)	Covered Use Case
R-AUTH-01	TC-AUTH-01, TC-AUTH-02	User Login and Authentication
R-AUTH-02	TC-AUTH-03, TC-AUTH-04	Role-Based Access Control
R-JOB-01	TC-JOB-01	Browse and View Jobs
R-JOB-02	TC-JOB-02	Employer Job Management
R-APP-01	TC-APP-01, TC-APP-02, TC-APP-03	Job Application and Application Processing
R-RES-01	TC-RES-01	Resume Management
R-INT-01	TC-INT-01, TC-INT-02	Interview Invitation and Response
R-NOT-01	TC-NOT-01, TC-NOT-02, TC-NOT-03	Notification Management
R-ADM-01	TC-ADM-01	Administrative Management
R-SUP-01	TC-SUP-01	Support Ticket Management
R-FBK-01	TC-FBK-01	Feedback Management

Table 7.6: Traceability of test cases to system requirements and use cases

7.3 Techniques Used for Test Generation

During the verification of several testing techniques were considered: CareerConnectBD. The test cases were chosen according to the functional requirements of the application in accordance with the requirements of the project. Role specific permissions, workflows, validation rules, authorization, boundaries and relationships between the key parts of the system.

7.3.1 Requirements-Based Testing

Test cases were obtained from the system functional requirements. Important operations such as authentication, job management, job application, Recruitment status updates, resume processing, interview processing, notifications, administrative and support workflows were This will be taken into account in the system level test scenarios.

7.3.2 Positive Testing

Valid input and authorized persons were checked using positive test scenarios. actions result in desired outcomes. Examples are successful login, or someone is about to log in. The ability to submit a valid job application, authorize job management, application These include the status updates feature, creating interviews, and submitting support tickets.

7.3.3 Negative Testing

Negative scenarios were thought out to ensure that invalid or unauthorized Appropriate rejection of operations. Some examples of invalid login: The credentials, the guest access to the protected routes, the unauthorized access to the role, Prevent duplicate job applications and application attempts.

7.3.4 Role-Based Access Testing

Each user role in CareerConnectBD is supported for access testing multiple roles. was employed for authorizing the boundaries between Admin, Employer and Job. Seeker users. Protected resources were tested for so that users can access. Only those functions to which they are delegated.

7.3.5 Integration Testing

Interactions between the related modules were tested with integration testing. An application for a job is an instance of a Job Seeker connecting to a Job Posting and its responsible Employer. Likewise, status of application changes and interview, Activities, notifications, support tickets and administrative workflows. involve multiple components of the system are interacting.

7.3.6 Build Verification

The frontend build process was also checked to make sure that the frontend assets are built. It might have been successfully compiled with the project's vite based build system. This

assisted to endorse frontend dependencies and application assets remained, Potentially able to work with the built-in system.

7.4 Test Results and Discussion

The primary workflows of the system were identified as being involved in the testing activities. CareerConnectBD operate in accordance with their goal. Authentication and role-based authorization safeguards limited system resources, Employers, and both roles. Job Seekers and Employers, as well as a separate workflow for role-specific modules. and Administrators.

The application workflows and job management are such that job opportunities can be: Handled and applications for recruiting the cycle. There's also a resume integration and interview operations in the system. Notifications, support tickets, feedback and administration management.

Notifications are one of the most used features of the CareerConnectBD platform and were given special attention. Have links to multiple business processes. Database notifications are used to The persistent notification mechanism is used for this. Realtime broadcast is an option. viewed as 'add-on' rather than essential to finish core business requests. So, if the optional fails or is not available, then failure or unavailability of the optional Not the underlying database from being used by the realtime broadcasting service. notification and main business function from completing.

Throughout development, regressions of existing functionality were detected via automated testing. The Laravel test suite was successfully run during the development phase to test important application use-cases and the production build for the frontend was checked via the project's Vite-based build process. Due to the academic nature of this project report, some test-count metrics are not reported unless they are taken directly from the final output of the repository test run.

The Goodness of the Test Suite is assessed.

The CareerConnectBD test suite covers all the relevant functional and integration paths of the application. It has the following major advantages: The authentication, authorization, role-based access, job seeker workflows are what it is. Employee workflows, employment applications, hiring procedures and notifications, Administering operations, and platform services.

The test suite also has negative scenarios for security-sensitive and business-critical operations. Testing unauthorized access to verify: Role-based security boundaries, duplicate application testing helps Ensure business rules are enforced and data consistency is maintained. Notification-related regression The optional external or realtime infrastructure is verified by testing further. Does not unnecessarily affect the core operation of the application.

The automation of feature testing and integration-based testing will significantly reduce the workload of testers. Automated feature testing and integration-based testing will drastically minimize testers' workload. Validation testing, Authorization testing, Regression testing and Frontend testing build verification offers a solid basis to assess system reliability.

However, no test suite can assure that defects are completely eliminated. Further testing for production may involve additional performance tests at higher volume. The primary areas that need testing are testing, cross browser compatibility testing, and accessibility testing. There is penetration testing, and extended user acceptance testing. These activities can enhance its confidence further when the system is ready for large scale. production deployment.

7.5 Sustainability Plan

CareerConnectBD's sustainability plan is centred around maintainability, Scalability and reliability, security, and platform longevity.

The backend is modular, based on Laravel, providing the ability to develop individual modules. The architectural and structural elements should be preserved and added where it is necessary for the functional areas to be preserved and extended without redesigning the entire. system. Separation of responsibilities between controllers, services and models. Components for validation, authorisation, and user interface supports sustainable practices of development.

Structured persistent storage for the is implemented using the relational MySQL database. The key stakeholders of the platform. Database migrations offer a well managed way to change the database structure. Mechanism to allow evolution of the database schema as its requirements change.

Security sustainability is provided by authentication, role-based, We will also implement authorization, request validation, protected routes, and framework-provided data security. We will also use authorization, request validation, protected routes, and framework-provided data security. security mechanisms. These controls should be continually monitored and evaluated. As new functionality gets added.

Automated regression testing helps in achieving long-term reliability by making it possible for To ensure that existing workflows are verified after changes by the 'developers'. Version control With Git and repository-based collaboration, further supports traceability, and Team development and integration of changes under control.

The front end architecture can be further developed during the project's A Vue.js frontend development with a responsive interface built using Bootstrap. This is a rewrite of the components and the build workflow of Vite. The components of the back-end and front-end should Continue to be modular in order to be able to introduce improvements in the interface without. Unnecessary changes to the main business logic.

The environment-specific configuration should be considered for deployment sustainability. Separate from Application source code. Optional infrastructure, This should be independent of critical business processes such as realtime broadcasting. operations so that the core operation doesn't stop because of temporary external-service failures platform unavailable.

We designed the application using modules, automated testing, controlled database evolution, and development. Provide secure access control, version control and deployment practices, CareerConnectBD can be continually improved and extended based on future Requirements for a career and recruitment platform.

Challenges and Limitations

CareerConnectBD development was accompanied by implementation and integration of the following activities: Multiple functional modules for Job Seekers, Employers and Administrators. The project team faced a number of challenges during its development. Technical and project management issues. Addressing these challenges gave them hands-on experience in building a full stack web app. development, database, security, testing, team collaboration, and system integration.

8.1 Technical Challenges

The key technical challenges were developing and integrating a multi-role authentication and authorization system. CareerConnectBD supports It includes a variety of users such as Job Seekers, Employers and Administrators. There are different routes, dashboards, resources to access for each role, operations. Proper authentication and authorization of users to access the resources. middleware had to be carefully implemented in order to get authorization for their roles, Access control policies, route protection and role-based access control mechanisms.

An additional challenge was the management of the relationships between the key database entities. The system has some data that is interrelated with users, roles, All of the above that are related to the job seeker. All of the above pertaining to the job seeker. activities, notifications, support tickets, feedback and administrative operations. Applicability of relationships and consistency of data across These modules needed to be carefully designed and developed in their database schema, migrations, models, etc. relationships, and validation.

The job application and recruitment process was also integrated. challenges. A Job Application is the process of linking a Job Seeker to a particular Job. its responsible Employer. Further operations (including application status) must maintain proper ownership of updates, and activities for interviews and authorization. Ensuring unauthorized access and invalid operations is prevented. Needed further authentication and security protocols.

One other major technical hurdle was notification management. Notifications are created when key business events occur to various roles. For instance, if an Employer requires a notification to be sent to it because it has a requirement for a child's placement. Submit an application, a Job Seeker may need to receive updates as to the status of an application, or action to be taken in case of an interview occurs. Making sure that notifications would

be sent to the right person. The notification system and the required integration was a crucial aspect that needed careful consideration. corresponding business workflows.

The project had difficulties also with optional realtime notification broadcasting. While developing, an unavailable local realtime The normal application requests might be interrupted by the broadcasting service. The notification architecture needed to make sure that persistent Database notifications and core business operations might continue even if the There was no way of broadcasting in real-time (optional). This enhanced the ability to withstand. The notification workflow was given by me.

File and file management were other challenges. The application Manages various user and company resources such as: Images, logos, and files related to resumes. Ensuring correct file The storage, retrieval, path management, validation and presentation across Various interfaces needed to be carefully implemented.

The challenge of integration of Front and Back end was also a major concern. The project uses Laravel for backend application logic and Vue.js for frontend development, In addition to Bootstrap (Responsive interface design) and Vite (Frontend). asset management. Ensuring that the backend routes, APIs or nodes are consistent with one another. Responds to server calls, front end elements, authentication state and role specific There is a need to coordinate the frontend and backend development of interfaces. processes.

A different challenge arose during development with regard to database migrations. As the system grew, the database schema adapted to new users and additional features were added. features and relationships. Migration ordering and duplicate schema definition, Correlation and synchronisation of application code and the MySQL database had Care must be taken in managing it, so as to keep it at a consistent development environment.

Testing multiple modules that are connected together was also challenging. Changes to authentication, authorization, database relationships, The previously mentioned notifications or shared application components could impact. completed functionality. The automated testing and regression testing were Thus, it is essential for development to look for unwanted side effects.

8.2 Project Management Challenges

CareerConnectBD has been created as a collaborative effort in an academic software project. One of the primary project management issues was the system's totality being split up. to be broken down into smaller manageable pieces, but without having separately developed parts. May later be combined into one viable programme.

The challenge of coordinating the team was especially critical as it involved coordinating between the frontend and backend teams. Many development activities might rely on common routes, data structures, and other shared resources. Authentication mechanisms, database entities and business needs. Changes by one team member can impact components being developed by other team members. another team member. Therefore, communication and coordination were essential and important, Indispensable during the process of development.

Version control and collaborative development added another practical. challenge. The source code along with support team was managed using Git and GitHub. collaboration.

When dealing with multiple development branches, it was necessary to exercise caution because of the risk of them interfering with each other. managing commits, sync, merges and conflicts. Stability of a project and integration of new features was a significant step in the development process.

The other hurdle was to handle the project scope. CareerConnectBD contains Multiple modules such as authentication, user and role management, These offer a variety of features, including Job Seeker functionality, Employer and company management, job management, and much more. Apply online, resume features, recruiting process etc., notification, help and feedback, reporting and administration. Completing these modules and integrating them into the academic programs available. The development period needed to be prioritized and developed according to milestones.

Project scheduling was also impacted by testing and debugging. Some issues appeared It was not until the various modules were combined that this was made possible. Therefore, more time is needed to be added was needed for regression testing, debugging, database synchronization, and proof of role-specific processes.

The project team responded to these challenges by breaking up the work into tasks and established milestones to help them stay on track. The development process involves based development, version control, repeated testing, debugging, and continuous. Use of completed modules in integration.

8.3 System Limitations

CareerConnectBD offers the essential functionality needed for a job board, but it lacks a few features.CareerConnectBD offers the basic functions that are required for a job board, but it has a few missing features. The current system is a web-based career and job management system, There are some limitations which can be improved by future development.

Currently, it's mostly a web-based application. If you're on Android, download the app for your mobile device, and if you're using an iOS device, you can use the web app. current implementation. Users therefore use the system mostly via supported web browsers.

The notification system is capable of persistent application notifications, and realtime broadcasting is considered as an optional enhancement. The availability The instant realtime updates can thus be dependent on the deployment environment. and setup and availability of realtime broadcasting. infrastructure.

The existing system is not able to handle all the possible integration options of Large size commercial recruitment platforms. Advanced third-party integrations, Additionally, enterprise-level, and large-scale external recruitment networks. Additional development of services may be needed.

The platform will need to be further optimized prior to deployment at a A very large production scale is likely. A large number of concurrent users, very large datasets, In addition to the above, an advanced caching strategy, distributed infrastructure, and large-scale are also features of this platform. Investigations in performance optimization go beyond the scope of the current academic research. implementation.

Although there are security and validation procedures included in the system, there is still a possibility of a production More security activities would be helpful in deployment, including: The company provides professional penetration testing, continuous security monitoring, infrastructure, and other services. The company offers professional penetration testing, continuous security monitoring, infrastructure and more services. Automated backup strategies, regular dependency security audits and hardening.

Information and recommendations for recruitment are also quality dependent on the accuracy and completeness of data provided by users and Employers. Incomplete The profiles, resumes, company information, or job descriptions may be shortened or condensed. The effectiveness of information given via the platform.

Some sophisticated aspects that could add value to a big scale recruitment site, A wide range of analytics, sophisticated machine learning matching, Third party service integrations as well as native mobile applications, can be viewed as areas to be improved in the future and not as integral parts of it. About the existing system.

8.4 Lessons Learned

The project team had the opportunity for hands-on experience in the development of CareerConnectBD. Experience creating and deploying a full blown, multi-role web application. For the first time the importance of establishing a new lesson was learned. Before implementing complex and interconnected systems, a clear system architecture is needed. features.

The project proved that the use of the appropriate database design is significant. Because Many modules rely on user-to-user, user-to-role, company-to-company, job-to-job, etc., relationships. Changes to the database include additions to applications, resumes and recruitment activities. structure can have significant effects throughout the application. Careful The planning of database relationships migrations thus became an. Therefore, planning of database relationships migrations became an. An integral component of development process.

Another key insight was that business logic should be separated from The duties of presenting and receiving requests. Organizing application Mapping logic to the right controllers, services, models, validation etc., Improves maintainability and complex workflows can be handled by middleware and policies. easier to comprehend and evaluate.

The team also learned hands-on experience in role based access control. A multi-role application cannot be authenticated using just authentication. Authorizations should be applied uniformly to routes and resources to ensure that where every user is only allowed to execute the operations allowed by the system.

Automated testing was useful for ongoing development. When new Regression tests were added and/or modified due to features added or changed. Aided in uncovering unexpected impacts on existing functionality. This showed that it is important to have a stable test suite. During the entire software development process.

A major takeaway from the notification implementation was that, the design needed to be done with great care. fault-tolerant integrations. No additional infrastructure is to be built where it is not necessary. Become critical business operations dependent.

Separating persistent More responsive, user-centric experience helped build a more user-centric experience by delivering database notifications from optional realtime delivery. resilient architecture.

This project also had experience of front-end and back-end integration. Using Laravel for the back end and Vue.js in front end development. The data flow, routes, user interfaces and required was necessary to be carefully coordinated. role-specific functionality. The team was informed of the importance of regular communication The collaboration between Frontend and Backend Developers is crucial for the successful implementation of a system. integration.

Control systems and how they work. Practical experience of using version control through Git and GitHub collaboration. Collaborative software development, control, branching, commits and merging. The team was given the lesson that it is important to implement changes in an organized way and to keep it going, It is an appealing idea to have a meaningful development history, and tests integrated code before thinking about it. a milestone complete.

Lastly, CareerConnectBD proved to be successful in its development by showcasing the following: complete software system is an iterative process. Requirements analysis, Design, implementation, testing, debugging and refinement may be required. As each module is added on, revisited. The learning from the experience of... This has helped the team to better grasp Understanding software engineering principles and building a real world full stack application. development.

Future Work

CareerConnectBD is a fully-integrated Web-based career and Job Seekers with Employers. A recruitment management platform that unites Job Seekers and Employers together. Administrators via a centralized approach. The current implementation and job seeking. It offers core functionalities for job search, posting and seeking. HR Application Management, Resume Management, Recruitment Workflows, Notifications, support, feedback, and administration. However, the system can be further enhanced in the future by introducing advanced New technologies, other integrations, and deployment on a large scale.

9.1 Proposed Enhancements

A number of improvements can be made in the future versions of CareerConnectBD to maximize user experience, recruitment efficiency, scalability and platform. intelligence.

9.1.1 Advanced AI-Based Job Recommendation

A new version of CareerConnectBD could incorporate a more sophisticated artificial intelligence. Recommendation system based on intelligence and machine learning. Such a system was able to decipher Job Seeker profiles, educational requirements, work experiences, and abilities employment background, preferred work fields, desired work location, skill sets, Previous applications activities to offer more customized occupation. recommendations.

Likewise, Employers can get intelligent recommendations for candidates. screens for potential applicants according to the needs of a job vacancy. This kind of functionality can enhance the effectiveness of the recruitment process. and candidate-selection process.

9.1.2 Advanced Resume Analysis

The current resume management capability can be enhanced with advanced resume analysis capabilities. Future versions can automatically analyse (when possible) Scanned resumes to find skills, education, work experience, and more professional information.

The system can also give tailored recommendations on how to enhance their resumes. Write content and recognize missing information according to type of position a Job Seeker will submit an application for. This improvement would assist the users in making more enhanced creations. Fill in and update professionally appropriate profiles.

9.1.3 Intelligent Job and Candidate Matching

A more elegantly designed matching mechanism could be put in place to determine the match of job seekers and available job opportunities. Matching criteria can include skills, education, professional, and character. Based on personal experience, job type, location and other factors.

The matching result can be supplied as a compatibility score or ranking. This will make it easier for Job Seekers to find relevant job opportunities. and assist Employers in the hiring process when choosing appropriate candidates.

9.1.4 Native Mobile Applications

The development of dedicated mobile applications for Android and iOS can be done in the future. This could be achieved with a native mobile app that allows Job Seekers and Employers to get the most out of the platform. with easier access to the platform from mobile devices.

Mobile apps may have other features like push notifications, Track job applications, save job alerts, reminders to attend interviews, and applications social media status updates, accessibility and engagement.

9.1.5 Advanced Communication Features

Future editions of CareerConnectBD could include improved communication There are both Job Seekers and Employers between the two. These features could Provide structured recruitment messaging, interview-related communication, and communication history.

There would need to be proper authorization and privacy controls to ensure that s. that communication is limited to relevant and authorized parties.

9.1.6 Video Interview Integration

Training and integration with external video conferencing systems could be added. To assist with the process of hiring remotely. Employers might arrange online Directly recruit, send out meeting information and interviews. workflow.

This feature could make the platform more convenient for organizing interviews and make the coordinating process easier. It is appropriate for recruiting from afar and internationally.

9.1.7 Advanced Analytics and Reporting

The reporting abilities for administrative and Employer can be enhanced by using the Reporting Enhancement Add-in. advanced analytics dashboards. Future analyses may be able to give information about work. New products, new trends in applications, recruitment activity, candidate engagement, and platform usage.

Employers might start to assess the effectiveness of their recruitment strategies with the help of recruitment analytics. Platform performance and activity through the Dashboard. Administrators could use the Dashboard to check Platform performance and ac-

tivity, and their job postings were provided in a separate tab. Performance and user activities.

9.1.8 Enable integration of email and SMS notifications.

The notification system can be extended by adding on other external communication channels. The following is a list of optional events that may be generated by important system events. Email / SMS notifications based on user preferences.

Examples of this include reminders to interviewees, status changes for important applications, This is also important for account security notifications, as well as other time sensitive recruitment. activities.

9.1.9 The integration of Social and Professional Platforms

Future enhancements include external professional and Social platforms as applicable. Such integrations might make profile sharing, professional identity verification, or import of selected images.sharing, professional identity verification, or import of selected images. Professional information with user's consent.

If it is necessary to integrate with external systems, then the proper API should be available, Authorisation, privacy protection, and adherence to the policies of the external service provider.

9.1.10 Enhanced Security Mechanisms

Although authentication and authorization are being applied in the current system, The application level security including validation, middleware, policies, etc. Future deployments of these mechanisms could result in added security, measures.

Possible enhancements include multi-factor authentication, enhanced login The data collection includes monitoring, security event logging, automated suspicious-activity detection, regular vulnerability assessments, rate limiting improvements, professional penetration testing.

9.1.11 Cloud Storage and Content Delivery.

To deploy the uploaded resources for large scale (resumes, profile), follow these steps: Images, company logos and other such things could be moved to managed cloud object storage. This may also be accomplished by using a Content Delivery Network (CDN) to improve the The performance of the delivery of static and media resources.

This approach could be useful for increasing the scalability, availability, managing backups, etc. Geographically distributed users and and resource delivery.

Using this approach, the performance and scalability of the application are optimized.

In future, large scale versions of CareerConnectBD may include other features such as:

9.1.12 Performance and Scalability Optimization

Adopting a caching layer, optimizing databases, processing queues in the background, Distributed storage, horizontally scalable application, load balancing servers.

These enhancements would make this platform capable of serving much more traffic. number of users, jobs, applications, notifications, and concurrent requests.

9.1.13 Progressive Web Application Support

CareerConnectBD could also be enhanced as a Progressive Web Application (PWA). This may be the application-like thing to happen on supported devices. Make improvements for users who regularly use the platform via their smartphones. Make improvements for users who often use the platform through their cell phones. mobile browsers.

Depending on future implementation needs, the selected functionality can be: Also be optimized for unreliable or limited network connectivity.

9.2 Future Scope of the Project

CareerConnectBD is not only about the current scope, but has the potential to reach out to a larger audience in the future. An online job portals management system. Further development of the platform could become a larger ecosystem of digital career and recruitment.

One possibility would be to diversify the site from conventional job opportunities to a more wide-ranging category. Career Development's recruitment into comprehensive Career Development. Additional modules should be added to include: Career guidance, assessment of professional skills, suggestions for training. Personalized career planning, international experience, and internships.

The platform might also have verified Employer and organization profiles. In order to build transparency and trust. There are suitable verification systems that could Help Job Seekers identify "real" Employers and minimize the risk of "phishing" by fake Employers. misleading job advertisements.

A further potential space would be a tie up with educational establishments and professional training organizations. Education and training providers including universities and training providers might Post internships, career events and other job related content on the platform Provide employment-related information to students and graduates.

CareerConnectBD could be scaled up to facilitate special careers' employment. There are various categories available such as internships, entry-level, remote, and more. There are opportunities for freelancing, and other kinds of professional involvement. Additional filtering and recommendation systems can assist users to discover Each student is given access to career opportunities according to his or her career preferences.

In the long term, there is the possibility of using anonymized and aggregated recruitment data, To create (in accordance with privacy and data protection laws and regulations). labor-market insights. This analysis may be used to identify new skills. Changes in requirements, popular job categories and recruitment trends.

This architecture can also be scaled for a large-scale cloud. deployment. These include managed databases, distributed application services and cloud. integration with various services and toolchains.automated deployment pipelines, various services and toolchains integration, storage, monitoring infrastructure. A scalable service for notifications might be useful for a larger production environment.

The security is being continually improved and the performance optimized, As a part of this, CareerConnectBD has embraced and implemented intelligent technologies responsibly, The potential to grow from the academic software project to the scalable. career and recruitment platform will be able to handle broader range of Organizations, job seekers and employers.

Conclusion

10.1 Summary of the Project

CareerConnectBD is a web based career and recruitment management platform. Created to give a centralized platform for Job Seekers, Employers, and other stakeholders and Administrators. The main goal of the project is to make the process of simplifying, organize recruitment process by combining job discovery, job posting, Application Management, Candidate Management, Resume related activities (including uploading, saving, printing, searching, etc.), Integrated interview workflows, notifications, support services, feedback and All administrative operations in one platform.

Administrative systems are role based and each group of users has a distinct role. Access to system functions are given to an appropriate number of users. Job Seekers can keep their profile and career information up-to-date, Research job openings, apply for jobs, and handle their applications. Involve in application activities, keep resumes related information and get involved in recruitment workflows. Employers are able to keep company information, Search, review and manage applications for jobs, publish and manage job opportunities. Access application statuses, and conduct recruitment activities. The Administrator is responsible for the monitoring and management of the platform. Activities based on the privileges of the system.

For the implementation of CareerConnectBD, Laravel was chosen as the main backend system. Laravel was used as the main system of the backend for the implementation of CareerConnectBD. Server-side application logic is written in PHP and using application framework. MySQL is used for relational data management, whereas Laravel Eloquent ORM is used to temporarily store the data. It is used for relational data management and Laravel Eloquent ORM is used to temporarily store the data. Handle application data and model relationships. The frontend development Supports the use of Vue.js for interactive user-interface elements and uses Bootstrap to create responsive web design and interface. Laravel Breeze and The role-based authorization mechanisms of the application support authentication. Access to protection of system resources, and control thereof.

The system is divided into several functional modules in the following areas; In addition to authentication and authorization, Job Seeker profile management, Employer will include a comprehensive suite of features for managing their profiles. and company profile management, job management, job application processing, Resume/Career Information Management: Recruitment and Interview-related information sharing, notifications, support tickets, feedback and more. administrative operations.

The interdependence of the various phases is an important element of the project. An integral part of the recruitment process. A Job Seeker can find an appropriate job and fill out an application, and the Employer can check out the application. Install and configure the status of the application. Then, recruitment-related activities can be carried out with ease. Continue using the existing interview and communication means. Through the notifications, users get relevant information generated by important events. system activities.

Mechanisms of support and feedback are also included in the system. Users can submit administrative oversight when help is needed, can handle and respond to such requests. The feedback mechanism is: users who have an extra way to give opinions or ideas regarding the platform.

Security and integrity of data were discussed throughout the implementation. The idea of defining authentication, as well as role-based access control, middleware, authorization policies, etc. validate requests, implement CSRF protection, password protection mechanisms, and Limited access to secure resources add to the security. of the application.

Various tests were made to validate the major functional workflows and system requirements. The testing process included key areas including MSA, MM, RBAC, and prof management. Business operations, application processes, recruitment activities and processes, Correspondence to notifications, and other modules that form the core. These test validated that the For each implemented component, they function based on the specified system requirements.

Overall, CareerConnectBD illustrates the design and implementation of an integrated recruitment management system based on the latest web application development technologies. The system is developed and tested on a local development environment (local-host) and offers a structure that can be built for the future production deployment on a compatible hosting environment.

10.2 Objectives Achieved

The primary problem faced by the development of CareerConnectBD was tackled. Project goals set for the project. The main goals accomplished by: the completed implementation are summarized below:

- **The development of a centralized recruitment platform:** A web-based platform was created for a centralized location of Job Seekers, Employers, Administrators and to manage major career and recruitment. Activities in an integrated system.
- **Role-Based User Management:** The system specifies separate roles and control mechanisms for access. Job Seekers, Employers and Administrators. Users can access functionalities As per their roles and permissions.
- **Authentication and Authorization:** Ensure that users are registered, logged in, logged out and password operations are secured, The use of protected routes and access were put in place. Middleware, policies, and role-based authorization mechanisms are used to restrict unauthorized Access to protected system resources.

- **A Job Seeker Profile Management system consists of the following elements:** Job Seekers are given the tools to keep personal and career related profile information that is necessary in job search. recruitment activities.
- **Employer and Company Profile Management:** Employers can keep up to date Employer and Company information, filling out the organizational context for job postings. recruitment activities.
- **Job Management:** The platform allows employers to develop and manage employment opportunities, Job Seekers are able to browse for information about jobs that is available according to the The functions offered by the system.
- **Apply Management:** Employers can post jobs and Job Seekers can apply for jobs. Can access and utilise applications related to their job posts. This sets up a process of applying for jobs between candidates and Employers.
- **Application Status Management:** Employers can change the status of the applications that are relevant to the recruitment, enabling Job Seekers to be provided with suitable information on progress of their applications.
- **Resume and Career Information Management:** The system supports the functionality of the management of resume related and Ensure professional information is provided to support Job Seekers keeping information. relevant to their employment activities.
- **Design a Recruitment and Interview Workflow:** Recruitment-related processes integrated into system allowing To ensure employers and Job Seekers can continue with recruitment activities that are relevant. Once an application has been made.
- **Notification Management:** A notification mechanism was put in place which informs relevant users of A series of changes produced by critical business processes. Persistent database notifications feature access to notification history, and the Optional realtime delivery mechanisms can be added to architecture to improve the user experience.
- **Support Ticket Management:** Users are able to file tickets with the help center using the website; authorized Administrative users are able to handle and respond to support requests. This offers a formal means of communication for addressing user problems.
- **Feedback Management:** A feedback mechanism was put in place to enable comment by users, Suggestions/Opinions about the platform, supporting future evaluation and improvement of the system.
- **Administrative Management:** Administrative functionality was added to facilitate at platform level. Monitoring and management activities as per the Administrator's authorized responsibilities.
- **Responsive and Interactive User Interface:** The system offers a responsive web front end that's accessible across supported devices. Frontend is interactive with Vue.js. To achieve this, the application employs a development method, and for responsive layout and interface, the application uses Bootstrap. styling.

- **Relational Database Design:** MySQL was used to implement a structured relational database. The database supports the various entities and relationships needed to manage users, and As part of this process, profiles, jobs, applications, recruitment activities, notifications, etc., are tracked. Support services, feedback and administration.
- **Security and Data Validation:** Adequate controls and validation were integrated to Ensure the integrity of data and protection of system resources. These mechanisms include authentication, authorization, middleware, policies, request validation, and CSRF protection.
- **System Testing and Validation:** The main functional modules and workflows of CareerConnectBD were pilot tested. To check the behavior of the anticipated system. A testing process identified the following: The challenges associated with the implementation and the reliability of the core. functionalities.

The design and implementation of an integrated Web based career and recruitment management platform were the core efforts of CareerConnectBD and achieved the primary objectives. The project illustrates software engineering principles, database design, web application development, role based security, modular architecture and system testing. The current project is implemented locally, but the approach will facilitate the possibility of future enhancements including ML based matching features, native mobile apps, better analytics, and production deployment as future work.

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Appendices

The appendices provide supplementary information related to the installation, operation, development, and testing of CareerConnectBD. The materials include a user manual, a local installation guide, repository information, an optional screenshot appendix (kept as placeholders), and a small set of sanitized sample data examples.

The screenshot displays the phpMyAdmin interface for the CareerConnectBD database. The left sidebar shows a tree view of the database structure, including tables like cache, cache_locks, companies, contact_messages, educations, email_logs, experiences, failed_jobs, feedback, interview_invitations, jobs, job_applications, job_batches, job_listings, job_seeker_profiles, job_seeker_skills, migrations, notifications, password_reset_tokens, resumes, resume_builders, roles, and saved_jobs. The main panel shows a list of tables with columns for Table, Action, Rows, Type, Collation, Size, and Overhead. The table list includes details such as the number of rows, the storage engine (InnoDB), the character set (utf8mb4), and the collation (unicode_ci).

Table	Action	Rows	Type	Collation	Size	Overhead
cache	Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_unicode_ci	32.0 Kib	-
cache_locks	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 Kib	-
companies	Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_unicode_ci	96.0 Kib	-
contact_messages	Browse Structure Search Insert Empty Drop	12	InnoDB	utf8mb4_unicode_ci	64.0 Kib	-
educations	Browse Structure Search Insert Empty Drop	12	InnoDB	utf8mb4_unicode_ci	64.0 Kib	-
email_logs	Browse Structure Search Insert Empty Drop	20	InnoDB	utf8mb4_unicode_ci	32.0 Kib	-
experiences	Browse Structure Search Insert Empty Drop	6	InnoDB	utf8mb4_unicode_ci	32.0 Kib	-
failed_jobs	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 Kib	-
feedback	Browse Structure Search Insert Empty Drop	16	InnoDB	utf8mb4_unicode_ci	64.0 Kib	-
interview_invitations	Browse Structure Search Insert Empty Drop	7	InnoDB	utf8mb4_unicode_ci	64.0 Kib	-
jobs	Browse Structure Search Insert Empty Drop	52	InnoDB	utf8mb4_unicode_ci	112.0 Kib	-
job_applications	Browse Structure Search Insert Empty Drop	22	InnoDB	utf8mb4_unicode_ci	64.0 Kib	-
job_batches	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	16.0 Kib	-
job_listings	Browse Structure Search Insert Empty Drop	22	InnoDB	utf8mb4_unicode_ci	48.0 Kib	-
job_seeker_profiles	Browse Structure Search Insert Empty Drop	7	InnoDB	utf8mb4_unicode_ci	32.0 Kib	-
job_seeker_skills	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	96.0 Kib	-
migrations	Browse Structure Search Insert Empty Drop	31	InnoDB	utf8mb4_unicode_ci	16.0 Kib	-
notifications	Browse Structure Search Insert Empty Drop	6	InnoDB	utf8mb4_unicode_ci	32.0 Kib	-
password_reset_tokens	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	16.0 Kib	-
resumes	Browse Structure Search Insert Empty Drop	7	InnoDB	utf8mb4_unicode_ci	64.0 Kib	-
resume_builders	Browse Structure Search Insert Empty Drop	7	InnoDB	utf8mb4_unicode_ci	32.0 Kib	-

Figure 1: CareerConnectBD MySQL Database Structure in phpMyAdmin

The following screenshot presents the MySQL database structure of the CareerConnectBD system as viewed through phpMyAdmin in the local development environment. It provides visual evidence of the database tables used to support the major modules of the application.

User Manual

This user manual provides basic instructions for using the major features of CareerConnectBD. The system supports three primary authenticated user roles: Job Seeker, Employer, and Administrator. Publicly accessible features may also be available to guest users without authentication.

A.1 Guest User

A guest user can access publicly available sections of CareerConnectBD without logging into the system.

A guest user can perform the following basic operations:

1. Open the CareerConnectBD website.
2. Browse publicly available job opportunities.
3. View the details of an available job.
4. Access the registration and login interfaces.
5. Register for an appropriate account type according to the available registration workflow.

A.2 Job Seeker User Manual

A Job Seeker uses CareerConnectBD to maintain career information, discover job opportunities, submit applications, and participate in recruitment activities.

A.2.1 Job Seeker Login

1. Open the CareerConnectBD login page.
2. Enter the registered email address.
3. Enter the correct password.
4. Click the login button.
5. After successful authentication, access the Job Seeker area according to the system workflow.

A.2.2 Manage Job Seeker Profile

1. Login using a Job Seeker account.
2. Open the Job Seeker profile section.
3. Create or edit the available personal and professional information.
4. Provide valid information in the required fields.
5. Save the changes.

A.2.3 Manage Education, Experience, and Skills

The Job Seeker can maintain career-related information through the corresponding sections of the system.

1. Login as a Job Seeker.
2. Open the required Education, Experience, or Skills section.
3. Add a new record or select an existing record for modification.
4. Provide valid information.
5. Save the record.
6. Update or remove existing information where the corresponding operation is available.

A.2.4 Browse and View Jobs

1. Open the available jobs section.
2. Browse the displayed job opportunities.
3. Select a job to view detailed information.
4. Review the available job requirements and related information.

A.2.5 Save a Job

1. Login as a Job Seeker.
2. Select a job of interest.
3. Use the available save-job option.
4. Open the Saved Jobs section to access saved opportunities.

A.2.6 Apply for a Job

1. Login as a Job Seeker.
2. Select an available job.
3. Open the job application interface.

4. Provide the required application information.
5. Select or provide the required resume information where applicable.
6. Submit the application.
7. Access the application section to review relevant application information.

A.2.7 Manage Resume

1. Login as a Job Seeker.
2. Open the Resume section.
3. Create or manage resume-related information.
4. Provide the required data.
5. Save the information.

A.2.8 Use Resume Builder

1. Login as a Job Seeker.
2. Open the Resume Builder section.
3. Create a new resume using the available interface.
4. Provide the required resume information.
5. Save the resume.
6. Use the available preview functionality to review the resume.
7. Edit the resume if modifications are required.
8. Use the available print or download functionality where required.

A.2.9 Manage Job Applications

1. Login as a Job Seeker.
2. Open the application management section.
3. Review submitted applications.
4. Open an application to view available details and status information.

A.2.10 Respond to Recruitment Activities

When an Employer performs an applicable interview or recruitment action, the Job Seeker can access the corresponding information through the available recruitment workflow.

The Job Seeker can review the relevant invitation and submit an available response according to the implemented workflow.

A.2.11 View Notifications

1. Login to the Job Seeker account.
2. Open the notification interface from the navigation area.
3. Review available notifications.
4. Open the relevant notification to access associated information where applicable.

A.3 Employer User Manual

Employers use CareerConnectBD to maintain company information, manage job opportunities, review applications, and perform recruitment-related activities.

A.3.1 Employer Login

1. Open the login page.
2. Enter valid Employer account credentials.
3. Submit the login form.
4. Access the Employer area after successful authentication.

A.3.2 Manage Company Information

1. Login as an Employer.
2. Open the company management section.
3. Create company information if required.
4. Enter valid organizational information.
5. Upload supported company-related media where applicable.
6. Save the company information.
7. Update the information when required.

A.3.3 Create and Manage Jobs

1. Login as an Employer.
2. Open the job management section.
3. Select the option to create a new job.
4. Enter the required job information.
5. Submit the job information.
6. Use the available management operations to edit or otherwise manage authorized job postings.

A.3.4 Review Job Applications

1. Login as an Employer.
2. Open the application management section.
3. Select an application associated with an authorized job.
4. Review the available applicant information.
5. Access the candidate's resume information where authorized and available.

A.3.5 Update Application Status

1. Open an authorized job application.
2. Review the applicant information.
3. Select the appropriate application status.
4. Save the status change.
5. The corresponding recruitment workflow and notification process are executed according to the system implementation.

A.3.6 Manage Interview Activities

1. Open an eligible candidate application.
2. Access the available interview-related functionality.
3. Provide the required interview information.
4. Submit the interview action.
5. Review candidate responses through the corresponding recruitment workflow where available.

A.3.7 View Employer Notifications

The Employer can use the notification interface to review relevant system events, including applicable recruitment and application-related activities.

A.4 Administrator User Manual

The Administrator is responsible for authorized platform-level monitoring and management operations.

A.4.1 Admin Login

1. Open the CareerConnectBD login page.
2. Enter valid Administrator credentials.
3. Submit the login form.

4. Access the protected Admin dashboard.

A.4.2 Manage Users

1. Login as an Administrator.
2. Open the User Management section.
3. Review available user information.
4. Create or update user information where authorized.
5. Manage supported user account status operations.
6. Perform other available user-management operations according to administrative authorization rules.

A.4.3 Monitor Platform Resources

The Administrator can access the available administrative sections for monitoring and managing supported platform resources, including users, Employers, companies, jobs, applications, reports, and other administrative information provided by the system.

A.4.4 Manage Support Tickets

1. Open the Support Ticket management section.
2. Review submitted support requests.
3. Select a ticket to view its details.
4. Provide an administrative response where appropriate.
5. Manage the ticket according to the available support workflow.

A.4.5 Review Feedback

1. Open the Feedback management section.
2. Review feedback submitted through the platform.
3. Use the information for platform evaluation and future improvement.

A.4.6 Access Reports and Settings

Authorized Administrators can access the available reporting and system settings functionality. These sections support platform monitoring and management of the configuration options provided by the current implementation.

Installation Guide

This section describes the general procedure for installing CareerConnectBD in a compatible local development environment.

B.1 System Requirements

A compatible development environment requires the following software:

- PHP 8.4
- Composer
- Laravel 13 (installed via Composer)
- MySQL server
- Node.js and NPM (for Vite-based frontend asset development)
- Git
- A compatible web server or the Laravel built-in development server

B.2 Installation Procedure

1. Clone the project repository using Git:

```
git clone https://github.com/mdtareqmiah/CareerConnectBD1.git
```

2. Navigate to the project directory:

```
cd CareerConnectBD1
```

3. Install PHP dependencies:

```
composer install
```

4. Create the environment configuration file from the provided example:

```
copy .env.example .env
```

On Linux or macOS, the corresponding command can be used:

```
cp .env.example .env
```

5. Generate the Laravel application key:

```
php artisan key:generate
```

6. Create a MySQL database for CareerConnectBD.
7. Configure the database connection in the `.env` file according to the local environment.

Example configuration:

```
DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=careerconnect
DB_USERNAME=<database-username>
DB_PASSWORD=<database-password>
```

8. Run database migrations and the required seeders:

```
php artisan migrate --seed
```

For a clean development database, the following command can be used only when existing data can safely be removed (it drops all tables and deletes existing data):

```
php artisan migrate:fresh --seed
```

9. Install frontend dependencies:

```
npm install
```

10. Start the frontend development environment where required:

```
npm run dev
```

11. Start the Laravel development server:

```
php artisan serve
```

12. Open the local application URL displayed by the Laravel development server in a supported web browser.

B.3 Frontend Production Build

To create optimized frontend assets for production, execute:

```
npm run build
```

B.4 Running Automated Tests

The Laravel automated test suite can be executed using:

```
php artisan test
```

The command output provides the test execution result for the currently checked-out source code.

B.5 Clearing Application Cache

If configuration, route, or cache-related issues occur during development, the Laravel optimization cache can be cleared using:

```
php artisan optimize:clear
```

B.6 Storage Configuration

If the application uses Laravel’s public storage mechanism for uploaded files, the symbolic link can be created using:

```
php artisan storage:link
```

Filesystem permissions must allow the web server and application to read/write the required storage paths in the configured environment.

B.7 Realtime Notification Configuration

Persistent database notifications can operate independently of optional realtime broadcasting.

If realtime notification broadcasting is enabled in a deployment environment, the corresponding broadcasting infrastructure and environment variables must be configured correctly. If realtime broadcasting is not required, it should remain disabled according to the application’s environment configuration.

Source Code Repository

The CareerConnectBD source code is maintained using Git for version control and GitHub for repository hosting and team collaboration. The project repository is available at:

<https://github.com/mdtareqmiah/CareerConnectBD1>

The repository contains the source code and project resources required for local development and maintenance. Sensitive information (passwords, API secrets, private keys, and environment-specific credentials) is not included in this report and is not intended to be committed to the public repository.

Sample Test Data

Sample test data was used during development to demonstrate and verify major workflows in a controlled environment. The examples below illustrate the type of entities typically created via seeders or manual testing. They are intentionally sanitized and do not include real credentials.

Entity	Example (sanitized)
Admin account	Role: Admin; Email format: <code>admin@example.test</code>
Employer account	Role: Employer; Email format: <code>employer@example.test</code>
Job Seeker account	Role: Job Seeker; Email format: <code>jobseeker@example.test</code>
Company profile	Example company name: <i>CareerConnectBD Demo Company</i>
Job posting	Example title: <i>Junior Web Developer</i> ; Location: <i>Sylhet (demo)</i>
Job Seeker profile	Example skills: <i>PHP, Laravel, JavaScript</i> ; Education/experience records as applicable
Application	A Job Seeker applies to a posted job and receives an application record
Application status	Example statuses used in the workflow (e.g., <i>Pending, Reviewed, Short-listed</i>)
Notifications	Database notifications generated for actions such as application submission and status updates
Support ticket / feedback	Short demo messages created for testing admin review workflows

For security and privacy reasons, the report does not include real passwords, API keys, or personally identifying information. Any demonstration accounts used during evaluation should be dedicated test accounts within a local development database. environment.